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# Cortical Orofacial Motor Representation in Old World Monkeys, Great Apes, and Humans

## I. Quantitative Analysis of Cytoarchitecture

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**Key Words**
Brain evolution · Comparative neuroanatomy · Primary motor cortex · Mammals · Primates · Great apes · Facial expression · Communication · Grey Level Index

**Abstract**
Social life in anthropoid primates is mediated by interindividual communication, involving movements of the orofacial muscles for the production of vocalization and gestural expression. Although phylogenetic diversity has been reported in the auditory and visual communication systems of primates, little is known about the comparative neuroanatomy that subserves orofacial movement. The current study reports results from quantitative image analysis of the region corresponding to orofacial representation of primary motor cortex (Brodmann's area 4) in several catarrhine primate species *(Macaca fascicularis, Papio anubis, Pongo pygmaeus, Gorilla gorilla, Pan troglodytes*, and *Homo sapiens)* using the Grey Level Index method. This cortical region has been implicated in the execution of skilled motor activities such as voluntary facial expression and human speech. Density profiles of the laminar distribution of Nissl-stained neuronal somata were acquired from high-resolution images to quantify cytoarchitectural patterns. Despite general similarity in these profiles across catarrhines, multivariate analysis showed that cytoarchitectural patterns of individuals were more similar within-species versus between-species. Compared to Old World monkeys, the orofacial representation of area 4 in great apes and humans was characterized by an increased relative thickness of layer III and overall lower cell volume densities, providing more neuropil space for interconnections. These phylogenetic differences in microstructure might provide an anatomical substrate for the evolution of greater volitional fine motor control of facial expressions in great apes and humans.



## Introduction

The primary motor cortex plays a central role in controlling orofacial movements of anthropoid primates as manifest by the disproportionate size of the representation of the tongue, jaws, and facial muscles in the motor somatotopic map [Grünbaum and Sherrington, 1903–04;

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Vogt and Vogt, 1907; Walker and Green, 1938; Bailey et al., 1950; Woolsey, 1958; Hoffman and Luschei, 1980; McGuinness et al., 1980; Huang et al., 1988; Murray and Sessle, 1992a; Fogassi et al., 1994]. Anthropoids have exceptional abilities to control the configuration of their facial muscles to produce subtle signal gradations in vocal and gestural communication, allowing for skillful mediation of interindividual relationships [van Hooff, 1962, 1967; Schultz, 1969; Chevalier-Skolnikoff, 1973, 1982; Preuschoft and van Hooff, 1995; Hauser, 1996]. Such communication systems are critical in navigating a complex social environment, where reproductive success depends upon manipulation of conspecifics in temporary alliances and long-term associations [Cheney et al., 1987; Byrne and Whiten, 1988].

Behavioral evidence indicates interspecific variation in the control of orofacial muscles among anthropoids, particularly in the domain of facial expression. Compared to Old World monkeys, facial displays of great apes and humans involve a greater range of lower face mobility [van Hooff, 1962; Andrew, 1963, 1965; Preuschoft and van Hooff, 1995; Chadwick-Jones, 1998] with more extreme extrusions of the lips [Chevalier-Skolnikoff, 1973]. In addition, great apes, but not monkeys, have been frequently observed to make novel nonemotional orofacial contortions, seemingly for the purpose of play and self-amusement [van Lawick-Goodall, 1968; Chevalier-Skolnikoff, 1976, 1982]. Moreover, although several species of anthropoids appear to have the capacity to voluntarily inhibit emotional vocalizations and facial expressions to deceive social partners [de Waal, 1982, 1986; Goodall, 1986; Byrne and Whiten, 1992], many researchers have argued that great apes and humans have superior skill at suppressing affective output for the purposes of tactical deception [Yerkes and Yerkes, 1929; Chevalier-Skolnikoff, 1976, 1982; Whiten and Byrne, 1988; Byrne and Whiten, 1992].

Although the selective pressure of social complexity has frequently been cited as a major cause for the evolution of intelligence [Humphrey, 1976; Byrne and Whiten, 1988; Harcourt, 1988; Byrne, 1995] and brain structure [Holloway, 1973; Armstrong, 1985; Dunbar, 1998] in primates, currently there are very few data available to link particular species-specific social behaviors of primates with distinct changes in neural organization. The current study, along with the companion study on chemoarchitecture [Sherwood et al., 2004], were designed to examine interspecific variation in the microstructural organization of the primary motor cortex, a cortical area that is centrally involved in the voluntary control of orofacial movements, and therefore might be an important contributor to phylogenetic differences in the communication systems of anthropoid primates.

The primary motor cortex of primates corresponds cytoarchitecturally to a region designated area 4 [Brodmann, 1903, 1908, 1909; Barbas and Pandya, 1987], FA [von Bonin and Bailey, 1947; Bailey et al., 1950], F1 [Matelli et al., 1985, 1991] or areas 4a, 4b, and 4c [Vogt and Vogt, 1919]. This cortical area is recognizable in all primates according to its distinctive cytoarchitectural appearance [von Bonin, 1949; Geyer et al., 1998, 2000; Rivara et al., 2003; Sherwood et al., 2003b], with a combination of prominent giant Betz cells in the lower portion of layer V, a predominance of pyramidal type cells, low cell density and overall large cellular sizes, absence of an inner granular layer IV, a diffuse border between layer VI and the subjacent white matter, and overall poorly laminated appearance. Although these qualitative cytoarchitectural similarities, in conjunction with functional properties, allow for the recognition of area 4 as homologous across primates, subtle interspecific differences might exist in the cytoarchitectural organization of this area which would have significance for its processing capacity. The principal aim of the current study was to characterize the cytoarchitectural organization of the region corresponding to orofacial representation in area 4 of several catarrhine species *(Macaca fascicularis, Papio anubis, Pongo pygmaeus, Gorilla gorilla, Pan troglodytes,* and *Homo sapiens)* using a quantitative image analysis method that enables direct interspecific comparison. In light of known variation in dexterity and volitional control of orofacial muscles among anthropoid primate species, we hypothesized that the microstructural composition of primary motor cortex would exhibit phylogenetic differences.

## Materials and Methods

*Specimens and Tissue Preparation*

Tissue was obtained from twelve individuals (3 *Macaca fascicularis*, 1 *Papio anubis*, 2 *Pongo pygmaeus*, 2 *Gorilla gorilla*, 2 *Pan troglodytes*, 2 *Homo sapiens*). Table 1 shows the age and sex distribution of the sample. Specimens of macaque and baboon monkeys were obtained from animals perfused transcardially with 4% paraformaldehyde [for details see Hof and Nimchinsky, 1992; Hof and Morrison, 1995; Hof et al., 1996] in the context of unrelated experiments. Briefly, the animals were deeply anesthetized with ketamine hydrochloride (25 mg/kg, i.m.) and pentobarbital sodium (20–35 mg/kg, i.v. as necessary), intubated, and mechanically ventilated. The chest was then opened, the heart exposed, and 1.5 ml of 0.1% sodium nitrate was injected into the left ventricle. The descending aorta was clamped, and the monkeys were perfused transcardially with cold 1%

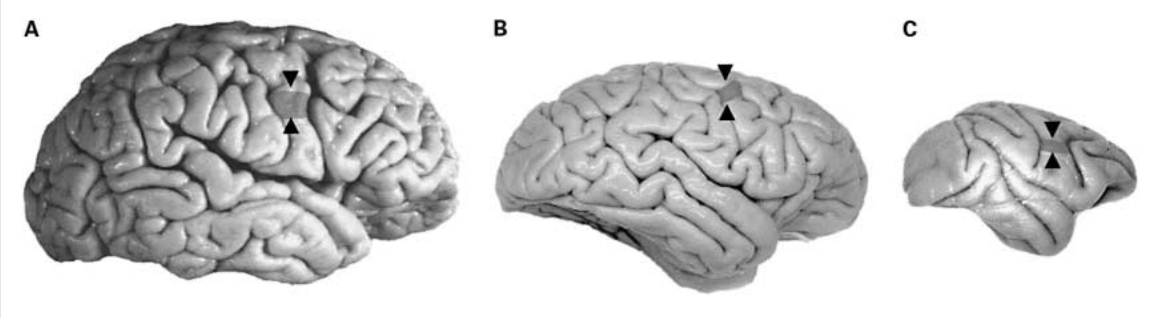



**Fig 1.** The location of orofacial representation in human (**A**), common chimpanzee (**B**), and long-tailed macaque (**C**). Tissue blocks were dissected from the precentral gyrus, central sulcus, and postcentral gyrus in the region of the cortical orofacial representation. Samples were taken from a location approximately 0.5 cm inferior to the level of the genu of the arcuate sulcus in Old World monkeys and roughly 1 cm inferior to the 'middle genu' of the central sulcus in great apes and humans. The sampling location is indicated by arrowheads. Note, the images are not to the same scale.

paraformaldehyde in phosphate buffer, pH 7.4, for 1 min followed by cold 4% paraformaldehyde for 10–12 min. The brains were then removed from the skull and postfixed for 6–7 h in 4% paraformaldehyde at 4°C. Great ape specimens were obtained from postmortem or terminally ill adult animals (31–38 years old) sacrificed for humane reasons. Immersion fixation of great ape brains in 10% neutral formalin began within 14 h of death and lasted for up to 9 days. Human brain specimens were obtained at autopsy within 7.5 h of death and immediately fixed by immersion in 10% neutral formalin for up to 9 days. The human brain samples derived from two neurologically and psychiatrically normal elderly individuals (a 65-year-old male and an 82-year-old female). Examination of clinical and behavioral records for nonhuman primates did not reveal any impairment of the motor system in any individual, thus it is unlikely that any abnormal conditions such as peripheral facial nerve injury would have affected our results. Although the human samples were obtained from elderly individuals, neither of these cases demonstrated any sensory-motor deficits on admission to the hospital and no such symptoms were recorded during their terminal illness. Moreover, the primary motor cortex is largely unaffected by normal aging [Tigges et al., 1990] and is generally spared even in late stages of Alzheimer's disease [Arnold et al., 1991]. Therefore, it is unlikely that the advanced age of the cases included in this study influenced the outcome of the analyses.

**Table 1.** Samples used in analyses of primary motor cortex

| Species | Sex | Age at death | Fixation | Postmortem interval |
|---|---|---|---|---|
| *Macaca fascicularis* | F | 13 | perfusion | – |
| *Macaca fascicularis* | M | 9 | perfusion | – |
| *Macaca fascicularis* | F | 6 | perfusion | – |
| *Papio anubis* | M | <10 | perfusion | – |
| *Pongo pygmaeus* | M | 38 | immersion | 8–14 h |
| *Pongo pygmaeus* | F | 31 | immersion | 8–14 h |
| *Gorilla gorilla* | M | 34 | immersion | 8–14 h |
| *Gorilla gorilla* | F | 32 | immersion | 8–14 h |
| *Pan troglodytes* | F | 38 | immersion | >1 h |
| *Pan troglodytes* | F | 37 | immersion | 8–14 h |
| *Homo sapiens* | M | 65 | immersion | 5 h |
| *Homo sapiens* | F | 82 | immersion | 7.5 h |

To ensure that the entire extent of area 4 was available, after fixation, 2–3 mm thick tissue blocks were dissected from the precentral gyrus, central sulcus, and postcentral gyrus in the region of the cortical orofacial representation (fig. 1). Only the right hemisphere was sampled in each brain. The right hemisphere was selected because production of many facial expressions has been shown to be right hemisphere dominant across catarrhines [Campbell, 1982; Hauser, 1993; Borod et al., 1997; Hauser and Akre, 2001; Fernández-Carriba et al., 2002], whereas speech, a highly specialized motor function, is localized to the left hemisphere in most humans [Kimura, 1993]. Published data from electrophysiological and functional imaging investigations in macaques [Lauer, 1952; Kwan et al., 1978; McGuinness et al., 1980; Humphrey, 1986; Huang et al., 1989], baboons [Craggs et al., 1976; Samulack et al., 1990; Waters et al., 1990], orang-utans [Leyton and Sherrington, 1917], gorillas [Leyton and Sherrington, 1917], chimpanzees [Grünbaum and Sherrington, 1903–04; Leyton and Sherrington, 1917; Hines, 1940; Dusser de Barenne et al., 1941; Bailey et al., 1950], and humans [Penfield and Boldrey, 1937; Penfield and Rasmussen, 1950; Lotze et al., 2000; Fox et al., 2001; Salmelin and Sams, 2002] were used to determine the approximate location of orofacial representation. Based on these studies, samples were dissected from a region located approximately 0.5 cm inferior to the level of the genu of the arcuate sulcus in Old World monkeys and from a location roughly 1 cm inferior to the 'middle genu' [Yousry et al., 1997] of the central sulcus in great apes and humans (fig. 1). It should be noted that, due to the lack of ante-

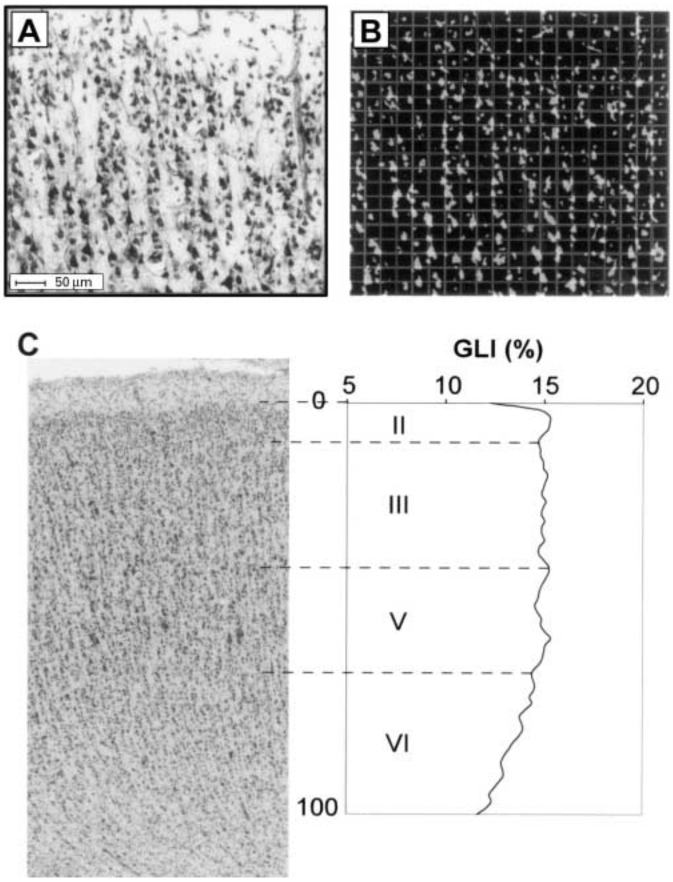



**Fig. 2.** Procedure for converting high-resolution histological images into GLI data. Each image frame (**A**) was converted to binary by adaptive thresholding, and was further subdivided into a grid of measuring fields (**B**). A GLI value was obtained for each measuring field. GLI profiles were then extracted and layer breakpoints were assigned. A GLI profile curve is shown superimposed over a high-resolution micrograph of the cortex of a chimpanzee specimen (**C**). The GLI profile represents laminar heterogeneity in cell volume density. The length of GLI profiles were standardized to a cortical depth of 100% in order to facilitate comparison among profiles derived from individuals of different regional cortical thicknesses. By drawing breakpoints between cortical layers, layer-wise mean GLIs can be extracted from the GLI profile curve.

mortem functional data in these individuals, the region of orofacial representation had to be estimated on the basis of species-typical somatotopic maps. Although every attempt was made to dissect blocks consistently from the superior part of orofacial representation, due to interindividual variation it is impossible to know with certainty the precise portion that was included in any individual sample. Nevertheless, according to intracortical microstimulation studies of primary motor cortex in macaques, the cortical representation of the mimetic facial muscles, jaw, and tongue overlap extensively and each muscle or movement is represented at multiple locations [McGuinness et al., 1980; Huang et al., 1988; Murray and Sessle, 1992a]. The tongue and the mimetic muscles, particularly zygomaticus and orbicularis oris, are represented across the greatest proportion of cortical territory [McGuinness et al., 1980; Huang et al., 1988; Murray and Sessle, 1992a]. Therefore, the region sampled most probably corresponds to the overlapping representation of tongue, jaws, and face.

Tissue blocks were cryoprotected by immersion in graded sucrose solutions (up to 30%) in PBS, pH 7.4, frozen on dry ice, and sectioned at 30 µm on a freezing sliding microtome perpendicular to the axis of the central sulcus. From this block, adjacent sections were collected for histological staining and stored in strict anatomical order. Sections not used for immediate staining were cryoprotected in a storage solution consisting of glycerol, ethylene glycol, $dH_2O$, and phosphate buffer (3:3:3:1 vol/vol) and stored in anatomical order at –20°C. Every 10th section (300 µm apart) from each block was mounted on chrom-alum subbed slides, stained for Nissl substance with thionin, and coverslipped with DPX. All sections were Nissl-stained together in order to ensure identical staining conditions across samples.

*Measurement of Grey Level Index*

Grey Level Index (GLI) was measured to quantify microstructural laminar organization. GLI represents the proportion of an area that is occupied by the projected profiles of all stained elements, and provides an estimate of the fraction of tissue that contains neuronal somata, glial nuclei, and endothelial nuclei versus neuropil [Wree et al., 1982; Schleicher and Zilles, 1990]. GLI values are highly correlated with the volume density occupied by neurons because glial and endothelial cell nuclei contribute only a very small proportion of the total volume [Wree et al., 1982].

GLI measurements were performed on high-resolution images of Nissl-stained sections. Five sections equally-spaced 300 µm apart were digitized and quantified from area 4 of each individual. A rectangular region of interest (ROI) that contained the entire extent of area 4 (see Results for discussion of boundary identification) on each section was digitized by using a microscope equipped with a motorized stage controlled by KS400 software (Carl Zeiss, Inc., Oberkochen, Germany). The entire ROI was imaged at low microscopic magnification with a Zeiss Planapo 6.3 × (1.25 N.A.) objective using an automated scanning procedure to acquire a montage of 8-bit grey scale image frames 524 × 524 µm in size and 512 × 512 pixels in spatial resolution. Each image frame was then converted to binary by adaptive thresholding, and was further subdivided into a grid of 32 × 32 measuring fields [Schleicher et al., 2000]. This sampling procedure resulted in a measuring field size of 16.4 × 16.4 µm and the entire cortical width was represented by at least 64 measuring fields. A GLI value was obtained for each measuring field and the resulting matrix of GLI data was stored for further processing (fig. 2).

GLI profiles were extracted from GLI-converted images. Profiles represent laminar heterogeneity in cell volume density (fig. 2). In order to generate GLI profiles, two contours were interactively drawn to define the cortical region for GLI profile extraction. The outer contour was defined as the boundary between layers I and II and the inner contour was drawn along the transition between layer VI and the white matter. The KS400 software then placed equidistant transverse measurement lines (200 µm apart) oriented vertical to the cortical layers and parallel to the cell columns [Schleicher et al., 2000]. On average, 21 ± 6 (mean ± SD) profiles were extracted from each section in Old World monkeys, 48 ± 8 in great apes, and 81 ± 5 in humans. The length of each profile was standardized to a cortical depth of 100% such that the border between layers I and II was located at 0% and the transition from layer VI to the white matter was located at 100%. Standardization of GLI profile length to a cortical depth of 100% allowed for comparison among profiles derived from individuals of different regional cortical thicknesses.

Each GLI profile was characterized quantitatively by extracting a set of 10 feature vectors based on central moments to describe the shape of the curve. The first feature is the mean GLI. The next four features are the profile's first four central moments (the mean cortical depth at the center of gravity of the area beneath the profile curve, standard deviation, the skewness, and the kurtosis of the frequency distribution) and the next five features correspond to the first derivative of the curve [Amunts et al., 1997; Schleicher et al., 2000]. These features reflect different aspects of laminar changes in cell volume distributions across the cortex, and therefore can be interpreted in terms of cytoarchitectural patterns [for details see Amunts et al., 2003]. For example, meany.o is a measure of the overall cortical mean cell volume fraction and meanx.o, the relative cortical depth at the center of gravity, describes the relationship of the GLI between supra- and infragranular layers such that low values indicate a higher GLI in supragranular layers compared to infragranular layers. Features were *z*-transformed in order to assign equal weight to each feature. Mean GLI profiles were calculated for each brain from the total population of profiles across all five sections through area 4.

In addition, the mean GLI value for each cortical layer (= layer-wise GLI) was calculated by subdividing the mean GLI profiles from each brain into segments corresponding to layers II, III, V, and VI [fig. 2; Amunts et al., 1995, 1997]. Break points between cortical layers were assigned to positions on the GLI profile by projecting the image of the histological section over a print-out of the mean GLI profile with a microscope drawing tube and matching maxima and minima of the profile to cytoarchitecturally-identified laminar boundaries [Semendeferi et al., 1998]. These layer-wise GLIs could then be compared between groups to elucidate further phylogenetic differences.

*Measurement of Relative Layer Widths*

Subsequent to defining laminar boundaries for layer-wise GLI analysis, the width of cortical layers was estimated from a series of three sections per individual. The GLI profile maxima and minima, as well as cytoarchitectural features [von Economo and Koskinas, 1925; von Bonin, 1949], were used to guide determination of layer boundaries. In general, laminar patterns were similar enough across species to use a consistent set of criteria in determining layer boundaries. Layer I is thin and mostly acellular aside from a few isolated Cajal-Retzius cells. Layer II contains densely packed small granular and pyramidal cells. The border between layer II and III can be identified by a sharp decrease in cell density and the appearance of more medium-sized pyramidal neurons as well as frequent stellate cells. Layer III is a relatively thick layer that can be divided into three sublayers (IIIa, IIIb, and IIIc) on the basis of increasing somatic size of pyramidal neurons and decreasing cellular density [von Economo and Koskinas, 1925]. As layer IV is not present in primary motor cortex [von Economo and Koskinas, 1925; von Bonin, 1949; Geyer et al., 2000], layer V is directly subjacent to sublayer IIIc. The transition from sublayer IIIc to layer V is marked by the absence of the dense medium and large pyramidal cells that characterize the bottom of layer III. Layer V can be divided into two sublayers (Va and Vb), with giant Betz cells located preferentially in its lower half [von Economo and Koskinas, 1925; Meyer, 1987; Rivara et al., 2003; Sherwood et al., 2003b], and generally lower cell density in sublayer Vb compared to sublayer Va. Betz cells and large pyramidal cells are not present in layer VI permitting a clear definition of the boundary between sublayer Vb and layer VI. Layer VI is a relatively thick layer that contains medium pyramidal and fusiform cells.

**Table 2.** Species data used in correlation analyses

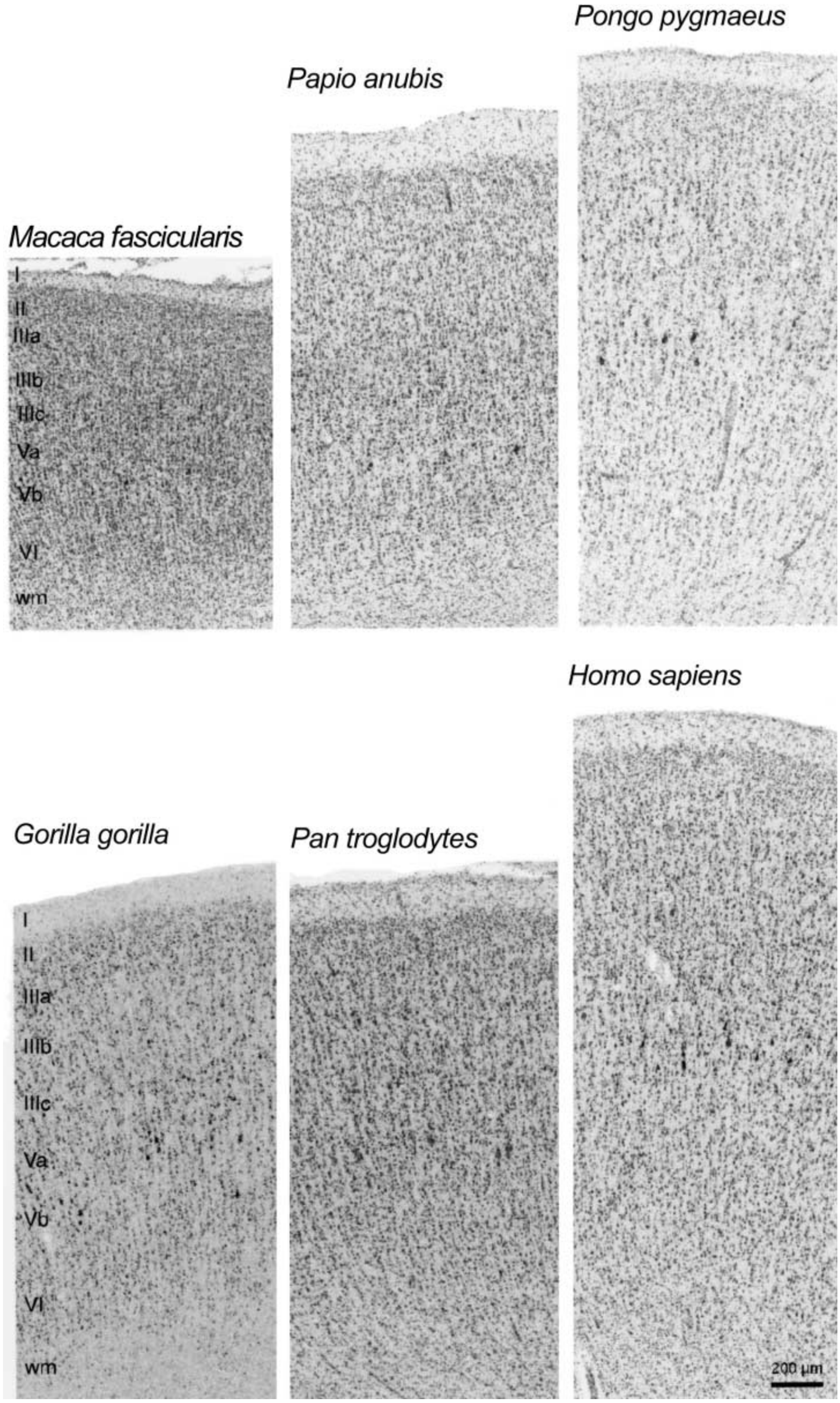
| Species | Body weight, g | Brain weight, g | Neocortex vol, $cm^3$ | EQ |
|---|---|---|---|---|
| *Macaca fascicularis* | 4,475 | 69.2 | – | 2.04 |
| *Papio anubis* | 19,200 | 175.1 | 140,142.0 | 1.72 |
| *Pongo pygmaeus* | 56,750 | 333.0 | 219,800.0 | 1.44 |
| *Gorilla gorilla* | 120,950 | 505.9 | 341,444.0 | 1.24 |
| *Pan troglodytes* | 52,750 | 405.0 | 291,592.0 | 1.85 |
| *Homo sapiens* | 60,000 | 1,250.0 | 1,006,525.0 | 5.19 |

In each section, measurements of the width of layers I, II, III, V, and VI were obtained from two locations along the anterior bank of the central sulcus, resulting in a total of six measurements of laminar width per individual. The regions of the sulcal fundus and gyral crown were avoided to prevent potential bias from distortion due to cortical folding at these sites. Laminar widths were converted to a percentage of the total cortical width for each measuring site and these six estimates of relative layer width were averaged to obtain a single value to represent the individual.

*Statistical Methods*

To improve statistical power, comparisons of laminar widths and layer-wise GLIs were made between individuals of the two catarrhine families, Cercopithecidae (i.e., Old World monkeys) and Hominidae (sensu Gray, 1825 – including great apes and humans). Prior to analysis, data within each group were tested for normality with the Shapiro-Wilk's W test and equality of variance was confirmed with Levene's test. Laminar and phylogenetic differences in layer-wise GLIs and layer widths were analyzed using a two-way analysis of variance (ANOVA) with repeated measures design. In this model, family was considered a main factor and layer was considered a within-subjects factor. The interaction effect of family × layer indicates the significance of phylogenetic differences in the distribution pattern of GLIs or widths among cortical layers. Follow-up multiple Student's t-test comparisons of layer values were made between families. Pearson product-moment correlation analysis was also used to explore relationships with species mean body weight, brain weight, neocortex volume, and encephalization quotient (EQ). Data from the literature were collected for species mean body weight, brain weight, and neocortex volume [table 2; Stephan et al., 1981; Harvey et al., 1987; Zilles and Rehkämper, 1988; Fleagle, 1999]. When separate mean brain and body weight values for males and females were available, they were averaged to obtain a species mean. EQ was determined for each species by taking the ratio of actual over expected brain weight. Expected brain weights were calculated from species mean body weights by using Martin's [1981] formula relating brain and body weights in a sample of 309 placental mammals [$\log_{10}$ brain wt. = 0.755 ($\log_{10}$ body wt.) + 1.774].

Several multivariate approaches were used to analyze GLI profiles. First, cytoarchitectural differences among species were quantified by calculating the Euclidean distances between profiles. Then, principal components analysis (PCA) was performed to summarize


Pongo pygmaeus
Papio anubis
Macaca fascicularis
I
II
IIIa
IIIb
IIIc
Va
Vb
VI
wm
Homo sapiens
Gorilla gorilla
Pan troglodytes
I
II
IIIa
IIIb
IIIc
Va
Vb
VI
wm
200 µm

the variance in the data and to determine the contribution of GLI features to phylogenetic differences in profile shape. Finally, discriminant function analysis was used to examine whether a priori classification of taxa was supported by GLI profile data and to determine which features were most essential to their differentiation. Statistical significance was set at $p < 0.05$. All statistical analyses were performed with Statistica version 6.0 (StatSoft, Inc., Tulsa, OK).

## Results

*Identification of Area 4 Boundaries*

Early cytoarchitectural maps of Old World monkeys and humans depicted primary motor cortex extending onto the convexity of the precentral gyrus [Brodmann, 1909; Vogt and Vogt, 1919]. More recent studies in humans and apes, however, have shown that, due to interindividual variation, sulcal landmarks cannot be used as reliable topographical criteria to localize precisely the boundaries of many cortical areas [Rademacher et al., 1993; Amunts et al., 1999, 2000; Geyer et al., 1999, 2000; Sherwood et al., 2003a], including the primary motor cortex [Geyer et al., 1996; White et al., 1997; Rademacher et al., 2001; Rivara et al., 2003]. Therefore, in the present study only cytoarchitectural criteria were used to identify the boundaries of the primary motor cortex (fig. 3). The posterior boundary of the primary motor cortex (between Brodmann's areas 3a and 4) was recognized by the appearance of a granular layer IV (in area 3a), an increasing cellular density in supragranular layers and a sharp division between white and grey matter [White et al., 1997]. This boundary occurs close to the fundus of the central sulcus in haplorhine primates [Sanides, 1970; Geyer et al., 1998, 2000; Rivara et al., 2003]. The anterior boundary of the primary motor cortex was more difficult to discern due to its gradual transition into Brodmann's area 6. Recently, it has become widely accepted that Betz cells in layer V persist into area 6 [Wise, 1985; Zilles, 1990; White et al., 1997; Geyer et al., 2000] making it inappropriate to determine this boundary on the basis of Brodmann's [1903, 1909] description of area 4 as the entirety of the precentral gigantocellular cortex. Nevertheless, area 6 can be distinguished from area 4 by the appearance of large, more elongated, and more densely packed pyramidal neurons appearing in the deep portion of layer III [Bailey and von Bonin, 1951; Zilles, 1990; Geyer et al., 1996; White et al., 1997]. Although there is some evidence that at least two architectural subdivisions exist within the regions of fore- and hindlimb representation in area 4 of anthropoids [Stepniewska et al., 1993; Geyer et al., 1996; Preuss et al., 1997; Binkofski et al., 2002], most researchers consider the region of orofacial representation to comprise a single functional and cytoarchitectural field [but see Preuss et al., 1997]. In the current study, no subdivisions were defined within area 4. The ROI defined for imaging and analysis corresponded to the cortex of the anterior bank of the central sulcus beginning at the fundus where the border between area 3a and area 4 occurred and extending anterior to the point where pyramidal neurons in layer III began to appear enlarged.

**Fig. 3.** Cytoarchitecture of the region of orofacial representation of primary motor cortex (Brodmann's area 4) in catarrhine primates. This cortical area is recognizable in all primates according to its distinctive cytoarchitectural appearance, including Betz cells in the lower portion of layer V, a predominance of pyramidal type cells, low cell density and overall large cellular sizes, absence of an inner granular layer IV, a diffuse border between layer VI and the subjacent white matter, and overall poorly laminated appearance.

*Relative Cortical Layer Widths*

Species mean relative cortical layer widths are shown in table 3 and figure 4. The relative width of layers in area 4 differed between families as revealed by the significant family × layer interaction effect ($F_{4,40} = 4.93$, $p = 0.003$). Although there was overlap in the range of values between some hominids and cercopithecids, follow-up t tests showed that layer III of hominids was relatively thicker than layer III of cercopithecids ($t_{10} = -2.86$, $p = 0.017$) and layer V of cercopithecids was relatively thicker than layer V of hominids ($t_{10} = 2.80$, $p = 0.019$). In contrast, layers I, II, and VI were homogeneous in relative width among species.

Correlation analysis determined whether differences in relative layer widths were related to overall size factors, such as body weight, brain weight, neocortex volume, or EQ. A negative correlation between the relative width of layer V and brain weight ($r = -0.894$, $p = 0.041$, $n = 6$) was the only relationship evident.

*Grey Level Index*

*Cortical Mean and Layer-Wise GLI Analysis.* Species mean whole cortical GLIs are shown in figure 5. The mean cortical GLI was significantly lower in hominids compared to cercopithecids ($t_{10} = 3.82$, $p = 0.003$). The mean cortical GLI of one chimpanzee individual, however, fell within the cercopithecid range. Among hominids there was a considerable amount of variation in GLIs (coefficient of variation = 19.1%). Gorillas had the lowest GLI (mean = 8.76) and chimpanzees had the highest GLI

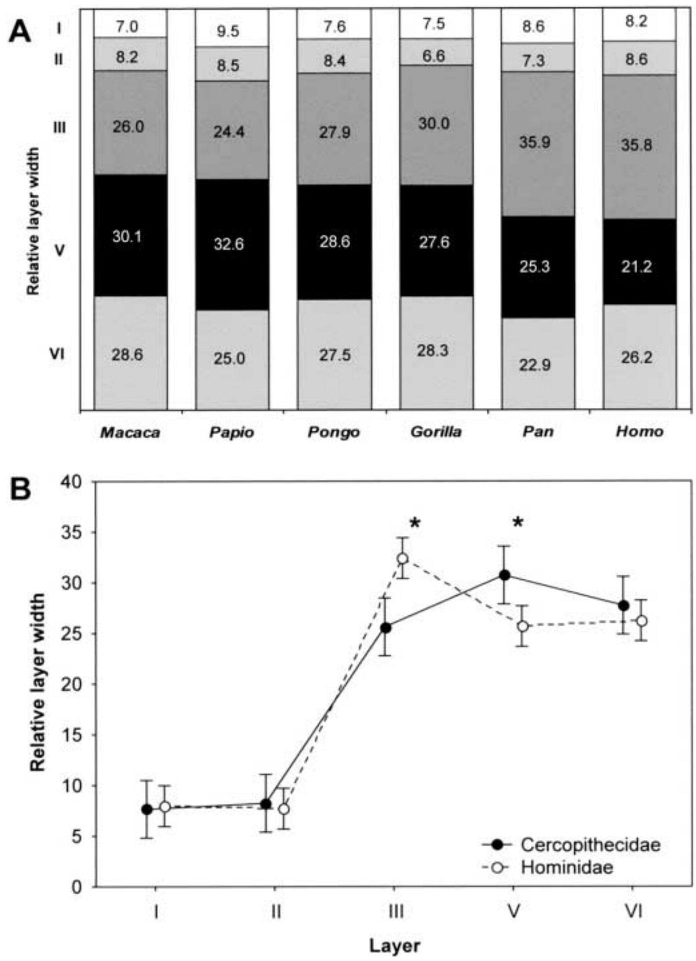



**Fig. 4.** Species mean relative layer widths. Each bar represents the proportion of the total cortical thickness occupied by each layer (**A**). Relative layer widths were statistically compared between families (**B**). Vertical bars denote 95% confidence intervals and asterisks indicate contrasts that were found to be significant by follow-up t test.

**Table 3.** Percentage of total cortical width represented by each cortical layer

| Species | n | % of total cortical width | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | | layer I | layer II | layer III | layer V | layer VI | layers I–III | layers V–VI |
| *Macaca fascicularis* | 3 | 7.0 | 8.2 | 26.0 | 30.1 | 28.6 | 34.2 | 58.7 |
| SEM | | 0.6 | 0.5 | 1.3 | 1.1 | 1.9 | 1.3 | 1.0 |
| *Papio anubis* | 1 | 9.5 | 8.5 | 24.4 | 32.6 | 25.0 | 32.8 | 57.6 |
| SEM | | – | – | – | – | – | – | – |
| *Pongo pygmaeus* | 2 | 7.6 | 8.4 | 27.9 | 28.6 | 27.5 | 36.3 | 56.1 |
| SEM | | 0.8 | 0.4 | 0.8 | 2.1 | 4.1 | 1.2 | 2.0 |
| *Gorilla gorilla* | 2 | 7.5 | 6.6 | 30.0 | 27.6 | 28.3 | 36.6 | 55.9 |
| SEM | | 0.5 | 0.2 | 0.5 | 0.8 | 0.9 | 0.3 | 0.2 |
| *Pan troglodytes* | 2 | 8.6 | 7.3 | 35.9 | 25.3 | 22.9 | 43.2 | 48.2 |
| SEM | | 0.6 | 0.6 | 2.6 | 0.5 | 3.4 | 3.3 | 3.9 |
| *Homo sapiens* | 2 | 8.2 | 8.6 | 35.8 | 21.2 | 26.2 | 44.4 | 47.4 |
| SEM | | 0.3 | 0.3 | 3.3 | 0.1 | 3.2 | 3.0 | 3.2 |

(mean = 13.19), with no apparent relationship between GLI and species mean brain size among hominids.

As a first approach to understanding the laminar pattern of phylogenetic differences in cytoarchitecture, GLIs were analyzed on a layer-wise basis. Table 4 and figure 5 show species mean layer-wise GLI measurements. There was a significant main effect of family on layer-wise GLIs ($F_{1,10} = 13.32$, $p = 0.004$), indicating that hominids (mean = 11.05, SD = 2.11) have lower overall GLIs than cercopithecids (mean = 15.40, SD = 1.05). Follow-up t-test comparisons of layer-wise GLIs showed that, compared to cercopithecids, hominids had lower GLIs for each individual cortical layer (all comparisons $p < 0.05$).

There was a significant effect of layer in the ANOVA model ($F_{3,30} = 19.67$, $p < 0.001$); Tukey HSD post hoc comparisons showed that GLIs were significantly lower in layers II and VI compared to layers III and V (all comparisons $p < 0.01$). Finally, the interaction effect of family × layer ($F_{3,30} = 3.57$, $p = 0.026$) was significant. Among cercopithecids, layers II/VI had lower GLIs than layers III/V (all comparisons $p < 0.05$, Tukey HSD post hoc), whereas among hominids, layer II had a relatively higher GLI making it statistically indistinguishable from other layers.

To determine if species differences in GLIs could be attributed to scaling effects, correlation analysis was performed on species mean GLIs, brain, and body size. Body weight was significantly correlated with GLI values in layer III ($r = -0.899$, $p = 0.038$, $n = 6$), layer V ($r = -0.925$, $p = 0.025$, $n = 6$), layer VI ($r = -0.920$, $p = 0.027$, $n = 6$), and the whole cortex ($r = -0.917$, $p = 0.028$, $n = 6$). The strength of the correlation with body weight was likely due to the pull of the *Gorilla* data point, however, as this species is both large-bodied and has low GLI values. Because the GLIs of gorillas appeared to be unusually low for a great ape, the data were reanalyzed for correlation after removing *Gorilla*. After removing gorillas there was no longer a significant correlation between GLIs and any brain or body size variable.

*GLI Profile Analysis.* GLI profiles were calculated for each individual to represent the characteristic laminar pattern of cell volume fraction distributions found in area 4. Figure 6 shows mean GLI profiles for each individual. Species mean GLI profiles are not illustrated as interindividual differences tend to smooth subtle details of the curve [Schleicher et al., 2000]. GLI profiles were quantitatively characterized by conversion into ten feature vectors (table 5), and these data were used in multivariate analyses. Euclidean distances were calculated between each individual GLI profile on the basis of feature vectors. The profiles of individuals from the same species tended to resemble one another more than profiles of individuals from different species as reflected by the lower Euclidean distances (fig. 7).

Multidimensional scaling of the ten feature vectors representing species mean GLI profiles are depicted in the Euclidean distance matrix (table 6) and two-dimensional plot (fig. 8). Surprisingly, Euclidean distances did not correspond to phylogenetic relatedness among species. For example, the distance between *Pongo* and *Homo* (0.6768) and *Macaca* and *Homo* (0.8766) were considerably smaller than the distance between *Pan* and *Homo* (1.1886). Thus, subtle differences in cytoarchitecture in this region do not appear to be directly determined by phylogenetic relationships.

**Table 4.** GLI values for each cortical layer

| Species | n | Layer II | Layer III | Layer V | Layer VI | Cortical mean |
|---|---|---|---|---|---|---|
| *Macaca fascicularis* | 3 | 14.06 | 16.08 | 16.13 | 14.26 | 15.58 |
| SEM | | 0.78 | 0.76 | 0.63 | 0.47 | 0.69 |
| *Papio anubis* | 1 | 14.11 | 14.28 | 15.76 | 14.34 | 14.85 |
| SEM | | – | – | – | – | – |
| *Pongo pygmaeus* | 2 | 10.34 | 11.09 | 10.96 | 9.56 | 10.61 |
| SEM | | 1.76 | 1.52 | 1.89 | 1.51 | 1.58 |
| *Gorilla gorilla* | 2 | 9.80 | 9.35 | 8.87 | 7.72 | 8.76 |
| SEM | | 0.61 | 0.46 | 0.61 | 0.62 | 0.63 |
| *Pan troglodytes* | 2 | 13.44 | 13.69 | 13.58 | 11.92 | 13.19 |
| SEM | | 1.17 | 1.21 | 1.14 | 1.09 | 1.15 |
| *Homo sapiens* | 2 | 10.29 | 11.83 | 12.21 | 11.13 | 11.65 |
| SEM | | 1.09 | 0.86 | 1.21 | 1.03 | 1.02 |

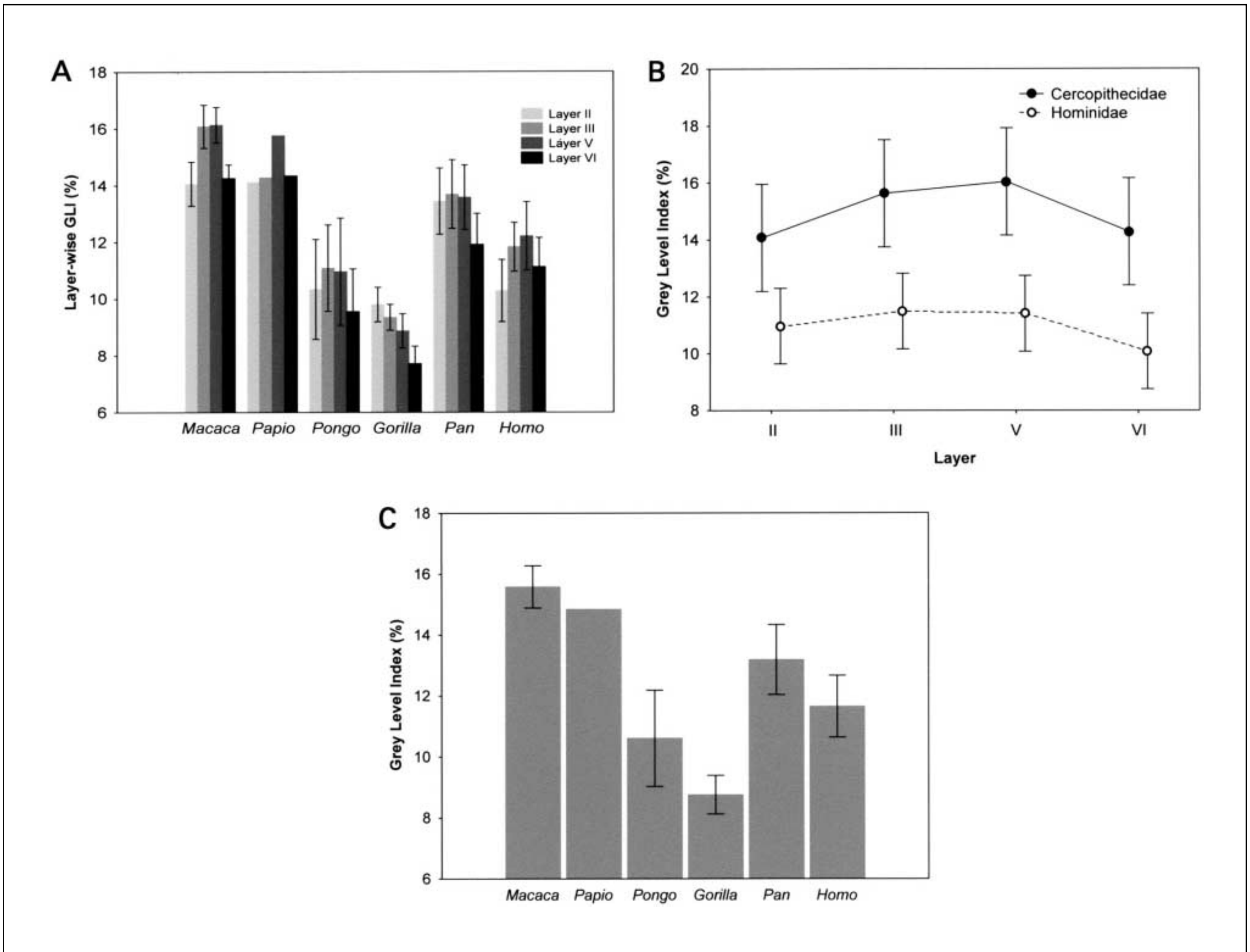
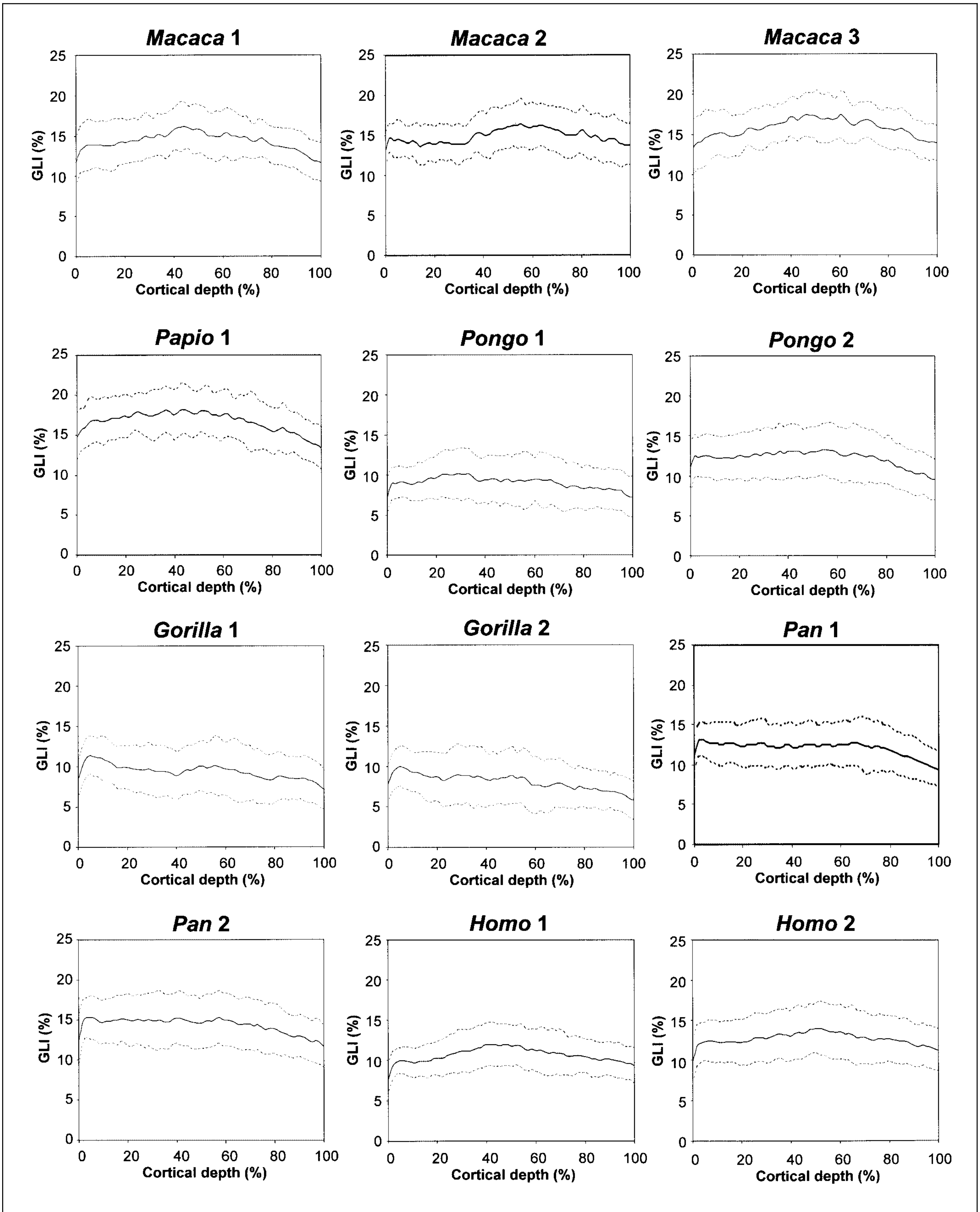



**Fig. 5.** Comparisons of species GLI values. A bar graph of layer-wise GLIs (mean ± SE) in area 4 is shown (**A**). Comparison of layer-wise GLI values between catarrhine families (**B**) revealed significantly lower overall GLIs in hominids (vertical bars denote 95% confidence intervals). Bar graphs of species mean whole cortex GLIs (mean ± SE) are shown (**C**). Although overall GLIs were lower in hominids compared to cercopithecids, among hominids brain size was not correlated with GLI.

Principal components analysis of the GLI feature data was performed to summarize the variance and to explore the features that best characterize phylogenetic groups. Ordination along nine axes was required to capture over 99% of the variance, but additional factors after factor 4 each explained less than 5% of the variance. Factors 1 and 2 combined summarized 79.4% of the overall variance. Factor 1 was the only axis that separated Old World monkeys and hominids (fig. 8). The variable contributions to factor 1 (table 7) show that Old World monkeys and hominids differed on a constellation of features including the center of gravity in the *x*-direction (meanx.o, meanx.d), the skewness (skew.o, skew.d), and the mean GLI (meany.o). Factor 2 which was mostly related to the standard deviation (sd.o, sd.d) and kurtosis (kurt.o, kurt.d), on the other hand, did not have much power to differentiate phylogenetic groups, besides separating the baboon from the

**Fig. 6.** Mean GLI profiles for each individual specimen with standard deviation indicated by dotted lines.


Macaca 1
Macaca 2
Macaca 3
Papio 1
Pongo 1
Pongo 2
Gorilla 1
Gorilla 2
Pan 1
Pan 2
Homo 1
Homo 2
GLI (%)
Cortical depth (%)
0
5
10
15
20
25
0
20
40
60
80
100

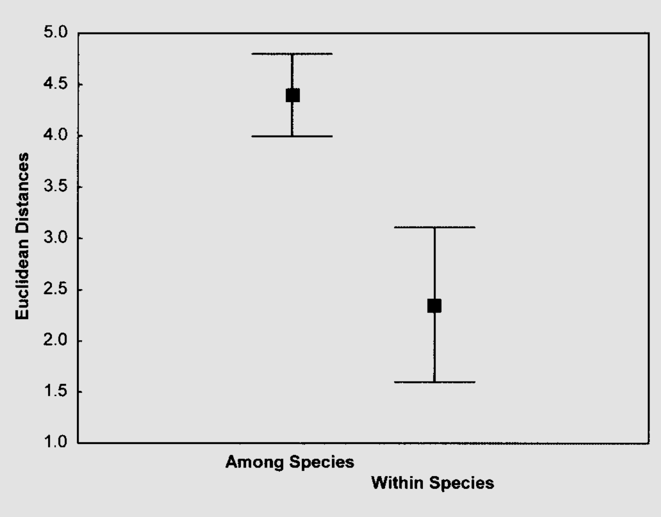



**Fig. 7.** Plot of Euclidean distances of GLI profile feature vectors among and within species (mean ± 95% confidence intervals).

rest of the catarrhines. Finally, factor 3 differentiated macaques from the other catarrhines and was related to mean GLI (meany.o, meany.d). Other than these patterns, however, there was overall poor separation among different species and higher-order phylogenetic groups along PCA axes, although individuals of the same species tended to cluster together.

Finally, stepwise discriminant function analysis was performed to assess whether GLI profile features can distinguish phylogenetic groups sufficiently to classify individuals to appropriate species and families. All ten feature vectors were entered; however, the reduced model had only six discriminant functions. Meanx.d, sd.d, skew.d, and kurt.d were not included in the reduced model because their contributions were redundant. The model correctly classified 100% of individuals to species and family, indicating that each taxonomic group is characterized by a unique cytoarchitectural pattern. According to

**Table 5.** Species mean values for 10 GLI profile feature vectors

| | *Macaca fascicularis* | *Papio anubis* | *Pongo pygmaeus* | *Gorilla gorilla* | *Pan troglodytes* | *Homo sapiens* |
|---|---|---|---|---|---|---|
| meany.o | 1.1103 | 0.8390 | −0.6807 | −1.3484 | 0.2476 | −0.3034 |
| meanx.o | 0.4560 | 1.3629 | −0.3290 | −1.5759 | −0.4176 | 0.9570 |
| sd.o | −0.9664 | 0.8894 | −0.4157 | 1.3694 | 0.6841 | −0.6330 |
| skew.o | −0.3711 | −1.7422 | 0.4265 | 1.4595 | 0.2545 | −0.7127 |
| kurt.o | 0.8608 | −0.3021 | 0.2162 | −0.9477 | −1.1899 | 0.7812 |
| meany.d | −1.4338 | −0.3579 | 0.6012 | 0.9407 | 0.1673 | 0.6205 |
| meanx.d | 0.5466 | 0.3909 | 0.4654 | −1.6839 | −0.5439 | 0.7471 |
| sd.d | 0.1008 | 2.5303 | −0.1575 | −0.9743 | 0.4816 | −0.7661 |
| skew.d | −0.5343 | −0.6195 | −0.1845 | 1.4597 | 0.3793 | −0.5433 |
| kurt.d | 0.2482 | −2.7462 | −0.0532 | 0.7372 | −0.3659 | 0.6827 |

Each GLI profile was characterized quantitatively by extracting a set of 10 feature vectors based on central moments to describe the shape of the curve. Features vectors are then *z*-transformed in order to assign equal weighting.

**Table 6.** Matrix of Euclidean distances between species mean GLI profiles

| | *Macaca fascicularis* | *Papio anubis* | *Pongo pygmaeus* | *Gorilla gorilla* | *Pan troglodytes* | *Homo sapiens* |
|---|---|---|---|---|---|---|
| *Macaca fascicularis* | 0.0000 | 1.5372 | 0.9801 | 1.9554 | 1.1806 | 0.8766 |
| *Papio anubis* | | 0.0000 | 1.6564 | 2.4300 | 1.4265 | 1.7244 |
| *Pongo pygmaeus* | | | 0.0000 | 1.2782 | 0.7821 | 0.6768 |
| *Gorilla gorilla* | | | | 0.0000 | 1.1077 | 1.7116 |
| *Pan troglodytes* | | | | | 0.0000 | 1.1886 |
| *Homo sapiens* | | | | | | 0.0000 |

Euclidean distances were calculated from the 10 GLI profile feature vectors in table 5.

the factor structure coefficients (i.e., the correlation between the variable in the model and the discriminant function), discrimination between families loaded mostly on meanx.o and meany.d.

## Discussion

The region corresponding to orofacial representation in the primary motor cortex of several catarrhine species was characterized using a quantitative cytoarchitectural method. Although primary motor cortex (Brodmann's area 4) could be easily recognized in all species on the basis of qualitative criteria (i.e., as frontal agranular cortex distinguished by gigantopyramidal Betz cells in layer V), quantitative image analysis revealed subtle differences in cytoarchitecture between phylogenetic groups that might have significant functional implications.

Functionally, the primary motor cortex is defined as the region of the frontal lobe from which simple movements of small sets of muscles can be evoked by low-amplitude electrical stimulation. It is necessary for the execution of voluntary movements [Evarts, 1966, 1968] and participates in the learning of motor skills [Rioult-Pedotti et al., 1998]. Individual neurons in the primary motor cortex encode the direction and amplitude of muscle force required to produce a specific movement, with changes in neuronal activity beginning about 100 ms prior to movement onset [Evarts, 1966, 1968]. Hence, the primary motor cortex is required for the control of kinematic and dynamic aspects of movement, whereas premotor areas trigger and guide movements on the basis of external and internal cues [Geyer et al., 2000].

**Table 7.** Variable contributions to principal components analysis of GLI profiles

| | Factor 1 (54.90%) | Factor 2 (24.45%) | Factor 3 (8.82%) | Factor 4 (5.94%) |
|---|---|---|---|---|
| meany.o | 0.1137 | 0.0105 | 0.2012 | 0.0763 |
| meanx.o | 0.1542 | 0.0002 | 0.0646 | 0.0394 |
| sd.o | 0.0693 | 0.2100 | 0.0143 | 0.0512 |
| skew.o | 0.1512 | 0.0100 | 0.0705 | 0.0248 |
| kurt.o | 0.0571 | 0.2113 | 0.0007 | 0.2371 |
| meany.d | 0.0871 | 0.0000 | 0.4842 | 0.0053 |
| meanx.d | 0.1424 | 0.0204 | 0.0880 | 0.1006 |
| sd.d | 0.0556 | 0.2464 | 0.0114 | 0.0850 |
| skew.d | 0.1342 | 0.0000 | 0.0550 | 0.3031 |
| kurt.d | 0.0351 | 0.2912 | 0.0099 | 0.0772 |

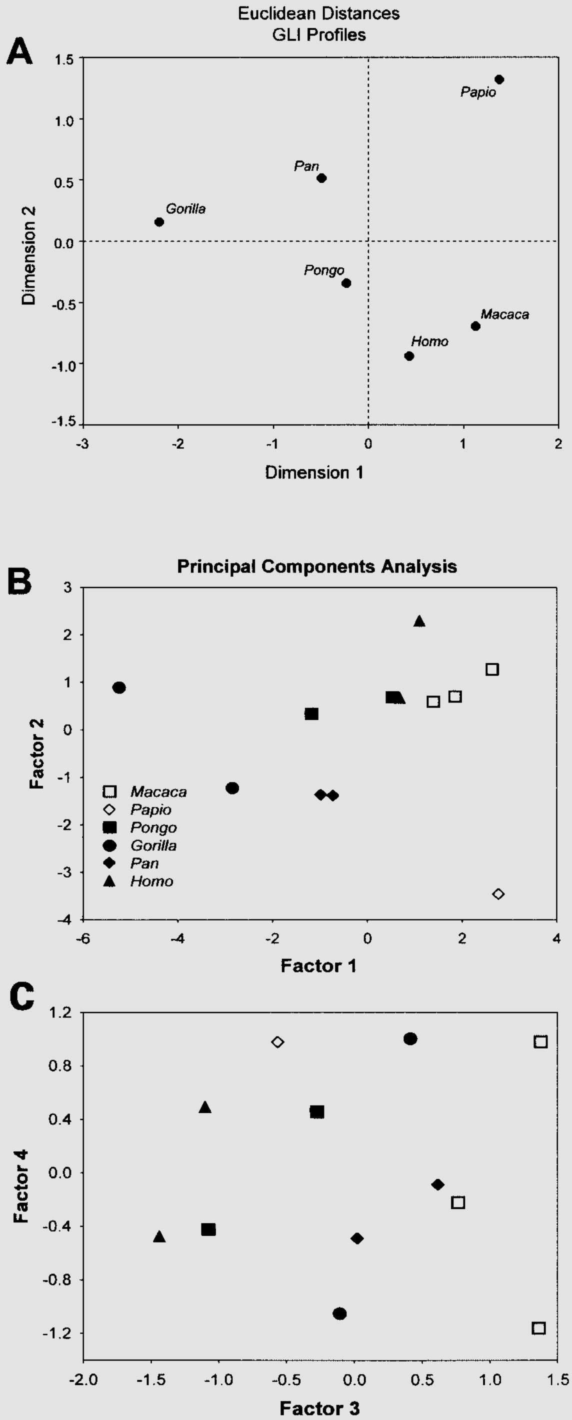



**Fig. 8.** Euclidean distance plot of species mean GLI profiles based on a multidimensional scaling procedure (see Materials and Methods for details) (**A**). Principal components plots summarizing GLI individual profile data are shown (**B, C**).

Electrical stimulation of primary motor cortex in owl monkeys [Preuss et al., 1996], squirrel monkeys [Hast and Milojevic, 1966], macaques [Hast et al., 1974; McGuinness et al., 1980; Gentilucci et al., 1988], and chimpanzees [Bailey et al., 1950] has been shown to evoke movements of the mouth, tongue, jaw, and vocal cords but few, if any, normal vocalizations or complete facial expressions. Results of electrical stimulation and lesion studies in primates implicate the orofacial region of primary motor cortex in swallowing and chewing [Green and Walker, 1938; Narita et al., 1999, 2002; Yamamura et al., 2002; Yao et al., 2002a, b]. In addition to these ingestive functions, the primary motor cortex also participates in circuits that regulate volitional control of learned motor patterns of the orofacial muscles [Murray and Sessle, 1992b, c; Lin et al., 1994a, b; Salmelin and Sams, 2002]. When the orofacial region of primary motor cortex is temporarily inactivated by cooling, for example, macaques have difficulty performing a trained motor task in which the tongue is protruded and pressed against a plate with a particular amount of force [Murray et al., 1991]. In anthropoid primates it has been observed that unilateral lesion confined to the precentral gyrus or the corticobulbar projection can produce contralateral deficits in the ability to produce voluntary movements of the lower face and tongue [Brodal, 1981; Rinn, 1984; Fridlund, 1994], but leave unimpaired spontaneous facial expressions evoked by emotional stimuli [Myers, 1969, 1976; Peters and Ploog, 1973; Brodal, 1981]. On a similar note, a recent neuroimaging study in humans found that the primary motor cortex was activated during posed voluntary smiling but not during spontaneous emotional smiling [Iwase et al., 2002]. In nonhuman primates, the primary motor cortex does not appear to be required for the production of vocalizations [Bailey et al., 1950; Hast and Milojevic, 1966; Myers, 1969; Hast et al., 1974; Sutton et al., 1974; Aitken, 1981; Jürgens et al., 1982; Kirzinger and Jürgens, 1982; Gentilucci et al., 1988]. Bilateral lesion of the region of orofacial representation in primary motor and premotor cortex, for example, does not significantly alter the acoustic structure of calls or the rate of calling in squirrel monkeys [Jürgens et al., 1982; Kirzinger and Jürgens, 1982] or macaques [Myers, 1969; Sutton et al., 1974; Aitken, 1981]. In contrast, the primary motor cortex is activated during speech and singing in humans [Salmelin and Sams, 2002] and bilateral destruction of the primary motor cortex in humans causes pseudobulbar palsy, a syndrome characterized by complete loss of voluntary control of linguistic phonation, with sparing of emotional utterances such as groaning, crying, and laughing [Alajouanine and Thurel, 1933; Foerster, 1936; Mariani et al., 1980; Groswasser et al., 1988; Mao et al., 1989]. Thus, interspecific differences in the execution of voluntary orofacial movements, such as non-emotional facial expressions and speech in humans, may be subserved by phylogenetic diversification in the structure and connections of the primary motor cortex.

Prior to the advent of computerized image analysis techniques, interspecific comparisons of the cytoarchitecture of primary motor cortex of primates were made on the basis of qualitative descriptions of cell morphology and laminar organization [Brodmann, 1903, 1905; Campbell, 1905; Mauss, 1908, 1911; Ariens Kappers et al., 1936; von Bonin, 1949] and analyses of differences in either volume [von Bonin, 1949; Svetukhina, 1956; Glezer, 1958] or neuron number [Campbell, 1905; Mayer, 1912; Lassek, 1940, 1941] in a limited number of species. An important limitation, however, of comparing species solely on the basis of regional volumes or cell number is that the underlying cytoarchitectural organization of the tissue might vary, and comparisons therefore are made of unequal units [Holloway, 1968; Armstrong et al., 1986]. Importantly, interspecific differences in the laminar microstructural organization of primary motor cortex could signify key differences in intrinsic circuits, afferent and efferent systems.

Before discussing our specific results, some methodological considerations deserve comment. There was a certain unavoidable degree of variability in tissue fixation methods in the sample. Old World monkey tissue was fixed by 4% paraformaldehyde perfusion, whereas the great ape and human tissue was fixed by immersion. This variability raises the question of how our results may be affected by shrinkage artifacts. Because the GLI method calculates a relative measure of the volume of neuronal somata per unit volume of brain tissue, global tissue shrinkage which affects both grey matter and white matter equally will not influence GLI values. Therefore, differential shrinkage between neuronal somata and neuropil is the only potential source of artifact that might bias our results. Kretschmann et al. [1982] found a 9% greater amount of shrinkage of grey matter versus white matter in human brains that had been fixed by immersion and embedded in paraffin. With respect to our data, the error due to histological artifact is likely to be less than that reported by Kretschmann et al. [1982] as the immersion-fixed specimens in our study were not subjected to further shrinkage from paraffin wax embedding. Considering that we found 33% lower GLI values in hominids compared to cercopithecids, biological variation rather than shrinkage artifact probably accounts for the majority of this difference. Most importantly, if we assume that differential shrinkage is not layer specific, any variability in histological artifact between phylogenetic groups would not influence the other features of the GLI profile curve as they are based on changes in GLI across cortical layers.

*Variation between the Catarrhine Families*

The present analysis revealed several interesting differences between catarrhine families in the cytoarchitecture of area 4. Discriminant function analysis was able to correctly assign 100% of individual GLI profiles to family, suggesting that there are consistent features that typify the cytoarchitectural organization of area 4 in these phylogenetic groups. The discrimination between families depended mostly on the mean GLI and the relative cortical depth at the center of gravity of the curve. These same features, along with skewness, were found to separate catarrhine families in the principal component analysis when no a priori information about phylogeny was included in the analysis. These results indicate that cercopithecids and hominids differ in the overall proportional of neuropil space as well as the laminar distribution of cell profiles. This was supported further by the finding that overall GLI values in area 4 were significantly lower in hominids compared to cercopithecids, indicating that hominids have relatively more neuropil space (fig. 9). Because there is not a correlation between the density of glial cells and brain size [Tower and Young, 1973b; Haug, 1987], interspecific differences in the GLI value are related mainly to the space occupied by neuronal profiles. The neuropil surrounding neuronal profiles consists of dendrites, axons, synapses, glial cell processes, blood vessels, and intercellular space [Schlaug et al., 1993; Pannese, 1994]. Therefore, the proportion of neuropil (i.e., the inverse of GLI) is an indirect indicator of the amount of synaptic interconnections within a region of cortex. Thus, lower GLI values among hominids corresponds to greater neuropil space between neurons, and indirectly, relatively more space occupied by neuronal processes and synapses, possibly providing enhanced integrative capacity.

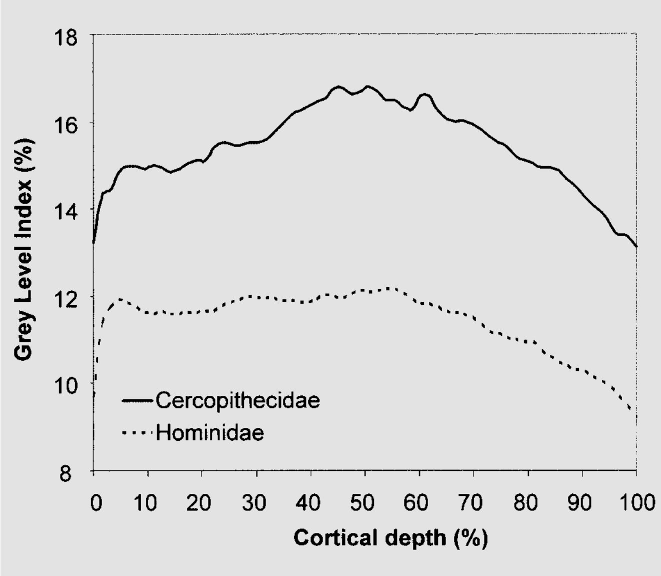



**Fig. 9.** Comparison of mean cercopithecid and mean hominid GLI profile curves.

Previous analyses of neuron densities, soma-volume fractions, and GLI in other cortical regions of mammals have reported a significant negative correlation between the space occupied by neurons and brain size [Shariff, 1953; Tower, 1954; Bok, 1959; Tower and Young, 1973a; Zilles et al., 1982; Armstrong et al., 1986; Haug, 1987; Prothero, 1997]. No such relationship was observed in the current study. Although Old World monkeys had significantly higher GLIs compared to hominids, the variability in GLI among hominids was not well predicted by species mean brain size. As an illustration of this point, although the human neocortex is approximately three times larger than the neocortex of great apes, GLIs were lower in orang-utans and gorillas than in humans. Haug [1987] showed that the neuronal soma volume fractions of chimpanzees, gorillas, humans, elephants, and whales are similar despite varying in brain volume by a factor of ten. Therefore, it seems that the correlation between neuropil space and brain size is found only in comparisons across a wide range of brain sizes. However, in comparisons across a more restricted range, (i.e., across great apes and humans), interspecific variability prevails over scaling trends. Furthermore, because human GLI and neuronal soma volume fraction [Haug, 1987] values fall within the range of great apes, we speculate that the neuropil fraction in the primate brain might reach an asymptote at the extreme of human brain size. This could occur due to network constraints to maintain local connectivity as a means to reduce the space occupied by wires and to minimize conduction delay with increased distance among interconnections [Ringo, 1991; Deacon, 1995; Chklovskii et al., 2002; Stepanyants et al., 2002].

There were several differences between catarrhine families in the laminar organization of area 4. Layer III of hominids was relatively thicker than in cercopithecids, and conversely, layer V was thicker in cercopithecids than in hominids. The relative expansion of layer III was particularly marked in chimpanzees and humans. Additionally, there was a significant interaction effect of family × layer for layer-wise GLIs that was associated with relatively higher neuronal volume density in layer II of hominids. In general, neurons in supragranular cortical layers are involved in corticocortical integrative processes and have axonal projections to other ipsi- and contralateral cortical area, whereas infragranular layers are comprised of neurons participating in corticofugal systems projecting to the spinal cord, brainstem, striatum, and thalamus [Jones and Wise, 1977; Szentágothai, 1983; Zilles, 1990]. Tract-tracing studies in monkeys have shown that the orofacial region of area 4 receives cortical inputs from the orbital cortex, frontal operculum, ventral premotor cortex, supplementary motor area, cingulate motor area, and parietal cortex [Leinonen and Nyman, 1979; Muakkassa and Strick, 1979; Godschalk et al., 1984, 1995; Huang et al., 1989; Morecraft and Van Hoesen, 1992, 1993; Darian-Smith et al., 1993; Luppino et al., 1993; Stepniewska et al., 1993; Morecraft et al., 1996; Tokuno et al., 1997]. Layers II and III of area 4, in particular, are the main target of projections from the somatosensory cortex which provides sensory feedback necessary for the execution of fine movements [Jones and Powell, 1968; Sloper, 1973; Jones et al., 1978; Vogt and Pandya, 1978; Sloper and Powell, 1979; Yumiya and Ghez, 1984; Ichikawa et al., 1985; Porter and Sakamoto, 1988; Kaneko et al., 1994]. Corticocortical output connections originating from neurons in layers II, III, and VI of area 4 project to the surrounding motor cortex, ventral premotor cortex, supplementary motor area, cingulate motor area, orbital cortex, insular cortex, primary and secondary somatosensory cortex, and inferoparietal cortex [Pandya and Vignolo, 1971; Künzle, 1978; Leichnetz, 1986; Alipour et al., 2002].

Because layer III is comprised of neurons that make corticocortical association connections, the finding of relatively thicker layer III in hominids suggests that the cortical orofacial representation of these species is involved in more complex processing that incorporates a wider array of inputs and output. Recent studies of motor learning in rats have shown that skilled reach training, but not simple motor activity, is associated with enhancement in dendritic branching [Withers and Greenough, 1989] and synaptogenesis [Kleim et al., 1996] in layers II and III of primary motor cortex. A substantial portion of the inputs to layer III of primary motor cortex are known to derive from the thalamus [Strick, 1973; Strick and Sterling, 1974] and primary somatosensory cortex [Porter and Sakamoto, 1988; Kaneko et al., 1994], implicating modification of this layer in sensorimotor integration and novel motor skill acquisition [Pavlides et al., 1993]. Thalamic afferent fibers targeting area 4 have been shown to originate from the ventral lateral nucleus pars oralis, the ventral posterolateral nucleus pars oralis, and the ventral lateral nucleus pars caudalis in macaques [Matelli et al., 1989] and chimpanzees [Walker, 1938]. These thalamic nuclei relay input from the cerebellum and basal ganglia [Asanuma et al., 1983; Ilinsky et al., 1985; Jones, 1985; Darian-Smith and Darian-Smith, 1993], carrying information related to the initiation, execution, and control of precise movements of the body.

Sensory feedback from superficial mechanoreceptors, relayed by the primary somatosensory cortex, may be particularly important in the performance of skilled orofacial movements. Superficial mechanoreceptive sensory input, for example, appears to be essential in the control of orofacial movements during human speech [Abbs and Cole, 1982] and in a trained tongue protrusion task in macaques [Murray and Sessle, 1992b, c; Lin et al., 1994a, b]. Thus, the relative expansion of layer III might subserve improved coordination and refinement of orofacial movements related to skilled movements in great apes and humans such as volitional facial expressions, feeding adaptations, and human speech.

In recent years, the relative thickness of layers in many different cortical areas have been measured for several primate species [Zilles et al., 1982; Zilles and Rehkämper, 1988; Semendeferi et al., 1998, 2001]. In prefrontal areas 10 and 13 of macaques and hominoids, the relative width of cortical layers did not appear to be related to brain size or phylogenetic lineage [Semendeferi et al., 1998, 2001]. A study of area 3, area 4, area 17, and area 41/42 in orangutans, gorillas, chimpanzees, and humans found that the relative widths of cortical layers were generally similar across great apes and humans [Zilles and Rehkämper, 1988]. This study unfortunately did not include non-hominid specimens, so catarrhine interfamily differences could not be determined. Interestingly, a relationship between relative laminar thickness in primary visual cortex (area 17) and activity pattern was found among prosimians [Zilles et al., 1982]. Diurnal species tended to have relatively thicker supragranular layers and a thinner granular layer compared to nocturnal species, whereas infragranular layers did not differ between these groups. In all, interspecific differences in laminar thickness are highly variable across cortical areas and appear to be dependent on a complex interaction between brain size scaling and functional architecture which is not yet fully understood.

*Distinctiveness of Species-Specific Cytoarchitecture*

Although within-species sample sizes were small, there was a remarkable amount of consistency in the shape of GLI profiles among individuals of the same species. The shape of GLI profiles was quantitatively summarized using ten feature vectors derived from the central moments of the curve. Euclidean distance analysis of the multivariate GLI feature data showed that the distance among individuals of the same species was significantly smaller than the distance among individuals of different species. In addition, ordination of GLI features along principal components axes resulted in clustering of individuals from the same species. Taken together, these data indicate that there are robust species-specific patterns of cytoarchitectural organization within the orofacial representation of area 4. This microstructure might form part of the neural basis for the distinctive orofacial movement that characterizes each catarrhine species. In general, facial expressions of great apes and humans are more varied and subtle compared to Old World monkeys [Chevalier-Skolnikoff, 1982]. However, even among great ape species there are differences in the use of facial expression. In one experiment that compared the facial expressiveness of captive chimpanzees, orang-utans, and gorillas just prior to feeding, it was found that chimpanzees used facial expressions most frequently and gorillas used them the least, with orang-utans intermediate [Maple, 1980]. Gorillas are generally considered to be the least overtly facially expressive of the great apes [Andrew, 1963; Hauser, 1996], although they have particularly expressive eye movements which might communicate subtle signals to coordinate group movements [M.M. Robbins and D. Morgan, pers. comm.]. Interspecific differences in non-communication orofacial skills might also explain some of the variation in cytoarchitecture. Orang-utans in the wild and captivity, for example, have been noted to exhibit particular facility at manipulating tools with their lips, teeth, and tongue during extractive foraging of insects and seeds [van Schaik et al., 1996; O'Malley and McGrew, 2000].

Surprisingly, Euclidean distance analysis of species mean GLI profiles showed that the overall cytoarchitectural configuration of area 4 was not related to phylogenetic relatedness. The GLI profile of humans, for example, was more similar to the macaque profile than to any of the African great apes. The human GLI curve differed from great apes in having a more sloping profile segment through the region corresponding to layer III. This difference signifies a relative increase in neuropil space within the superficial portion of layer III of humans that is not present in African great apes. We speculate that this modification of cytoarchitectural organization in the orofacial representation of humans could be related to increased associative connections in support of the production of prosody and spoken language. This hypothesis, however, awaits further verification by investigating the cytoarchitecture of other non-orofacial regions of area 4.

## Conclusion

The results of this study provide additional insight into the coevolution of the primary motor cortex and orofacial behavior of catarrhines. A comparative quantitative approach to cortical microstructure in the region of orofacial representation of primary motor cortex revealed several patterns of variation among species. The distinctiveness of species-typical cytoarchitectural organization might represent an underlying neural basis for the uniqueness of each species' mode of interindividual communication. Phylogenetic differences in microstructure of this brain region could provide an anatomical substrate for the evolution of the greater volitional and fine motor control involved in the facial expressions of great apes and humans.

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## References


Abbs JH, Cole KJ (1982) Consideration of bulbar and suprabulbar afferent influences upon speech motor coordination and programming. In: Speech Motor Control (Grillner S, Lindblom B, Lubker J, Persson A, eds), pp 159–186. Oxford, UK: Pergamon.

Aitken PG (1981) Cortical control of conditioned and spontaneous vocal behavior in rhesus monkeys. Brain Lang 13:171–184.

Alajouanine T, Thurel R (1933) La diplégie faciale cérébrale forme corticale de la paralysie pseudobulbaire. Rev Neurol 60:441–458.

Alipour M, Chen Y, Jürgens U (2002) Anterograde projections of the motorcortical tongue area in the saddle-back tamarin *(Saguinus fuscicollis)*. Brain Behav Evol 60:101–116.

Amunts K, Istomin V, Schleicher A, Zilles K (1995) Postnatal development of the human primary motor cortex: A quantitative cytoarchitectonic analysis. Anat Embryol (Berl) 192:557–571.

Amunts K, Schmidt-Passos F, Schleicher A, Zilles K (1997) Postnatal development of interhemispheric asymmetry in the cytoarchitecture of human area 4. Anat Embryol (Berl) 196:393–402.

Amunts K, Schleicher A, Ditterich A, Zilles K (2003) Broca's region: Cytoarchitectonic asymmetry and developmental changes. J Comp Neurol 465:72–89.

Amunts K, Malikovic A, Mohlberg H, Schormann T, Zilles K (2000) Brodmann's area 17 and 18 brought into stereotaxic space – Where and how variable? Neuroimage 11:66–84.

Amunts K, Schleicher A, Bürgel U, Mohlberg H, Uylings HB, Zilles K (1999) Broca's region revisited: Cytoarchitecture and intersubject variability. J Comp Neurol 412:319–341.

Andrew RJ (1963) Evolution of facial expression. Science 142:1034–1041.

Andrew RJ (1965) The origins of facial expression. Sci Am 213:88–94.

Ariens Kappers CU, Huber GC, Crosby EC (1936) The Comparative Anatomy of the Nervous System of Vertebrates, Including Man. Volume I. New York: Macmillan Company.

Armstrong E (1985) Enlarged limbic structures in the human brain: The anterior thalamus and medial mammillary body. Brain Res 362:394–397.

Armstrong E, Zilles K, Schlaug G, Schleicher A (1986) Comparative aspects of the primate posterior cingulate cortex. J Comp Neurol 253: 539–548.

Arnold SE, Hyman BT, Flory J, Damasio AR, Van Hoesen GW (1991) The topographical and neuroanatomical distribution of neurofibrillary tangles and neuritic plaques in the cerebral cortex of patients with Alzheimer's disease. Cereb Cortex 1:103–116.

Asanuma C, Thach WT, Jones EG (1983) Distribution of cerebellar terminations and their relation to other afferent terminations in the ventral lateral thalamic region of the monkey. Brain Res 286:237–265.

Bailey P, von Bonin G (1951) The Isocortex of Man. Urbana, IL: University of Illinois Press.

Bailey P, von Bonin G, McCulloch WS (1950) The Isocortex of the Chimpanzee. Urbana, IL: University of Illinois Press.

Barbas H, Pandya DN (1987) Architecture and frontal cortical connections of the premotor cortex (area 6) in the rhesus monkey. J Comp Neurol 256:211–228.

Binkofski F, Fink GR, Geyer S, Buccino G, Gruber O, Shah NJ, Taylor JG, Seitz RJ, Zilles K, Freund HJ (2002) Neural activity in human primary motor cortex areas 4a and 4p is modulated differentially by attention to action. J Neurophysiol 88:514–519.

Bok ST (1959) Histonomy of the Cerebral Cortex. Princeton, NJ: Van Nostrand-Reinhold.

Borod JC, Haywood CS, Koff E (1997) Neuropsychological aspects of facial asymmetry during emotional expression: A review of the normal adult literature. Neuropsychol Rev 7:41–60.

Brodal A (1981) Neurological Anatomy: In Relation to Clinical Medicine. New York: Oxford University Press.

Brodmann K (1903) Beiträge zur histologischen Lokalisation der Grosshirnrinde. Erste Mitteilung: Die Regio Rolandica. J Psychol Neurol 2: 79–107.

Brodmann K (1905) Beiträge zur histologischen Lokalisation der Grosshirnrinde. Dritte Mitteilung. Die Rindenfelder der niederen Affen. J Psychol Neurol 4:177–226.

Brodmann K (1908) Beiträge zur histologischen Lokalisation der Grosshirnrinde. Sechste Mitteilung: Die Cortexgliederung des Menschen. J Psychol Neurol 10:231–246.

Brodmann K (1909) Vergleichende Lokalisationslehre der Grosshirnrinde in ihren Prinzipien dargestellt auf Grund des Zellenbaues. Leipzig: Barth.

Byrne RW (1995) The Thinking Ape: Evolutionary Origins of Intelligence. Oxford, UK: Oxford University Press.

Byrne RW, Whiten A (1988) Machiavellian Intelligence: Social Expertise and the Evolution of Intellect in Monkeys, Apes, and Humans. Oxford, UK: Clarendon Press.

Byrne RW, Whiten A (1992) Cognitive evolution in primates: Evidence from tactical deception. Man 27:609–627.

Campbell AW (1905) Histological Studies on the Localization of Cerebral Function. London: Cambridge University Press.

Campbell R (1982) The lateralization of emotion: A critical review. Int J Psychol 17:211–229.

Chadwick-Jones J (1998) Developing a Social Psychology of Monkeys and Apes. East Sussex: Psychology Press.

Cheney DL, Seyfarth RM, Smuts BB, Wrangham RW (1987) The study of primate societies. In: Primate Societies (Smuts BB, Cheney DL, Seyfarth RM, Wrangham RW, Struhsaker TT, eds), pp 1–8. Chicago, IL: University of Chicago Press.

Chevalier-Skolnikoff S (1973) Facial expression of emotion in nonhuman primates. In: Darwin and Facial Expression (Ekman P, ed), pp 11–90. New York: Academic Press.

Chevalier-Skolnikoff S (1976) The ontogeny of primate intelligence and its implications for communicative potential: A preliminary report. Ann N Y Acad Sci 280:173–211.

Chevalier-Skolnikoff S (1982) A cognitive analysis of facial behavior in Old World monkeys, apes, and human beings. In: Primate Communication (Snowdon CT, Brown CH, Petersen MR, eds), pp 303–368. Cambridge, UK: Cambridge University Press.

Chklovskii DB, Schikorski T, Stevens CF (2002) Wiring optimization in cortical circuits. Neuron 34:341–347.

Craggs MD, Rushton DN, Clayton DG (1976) The stability of the electrical stimulation map of the motor cortex of the anesthetized baboon. Brain 99:575–600.

Darian-Smith C, Darian-Smith I (1993) Thalamic projections to areas 3a, 3b, and 4 in the sensorimotor cortex of the mature and infant macaque monkey. J Comp Neurol 335:173–199.

Darian-Smith C, Darian-Smith I, Burman K, Ratcliffe N (1993) Ipsilateral cortical projections to areas 3a, 3b, and 4 in the macaque monkey. J Comp Neurol 335:200–213.

de Waal FBM (1982) Chimpanzee Politics. London: Jonathan Cape.

de Waal FBM (1986) Class structure in a rhesus monkey group: the interplay between dominance and tolerance. Anim Behav 34:1033–1040.

Deacon TW (1995) On telling growth from parcellation in brain evolution. In: Behavioural Brain Research in Naturalistic and Semi-naturalistic Settings (Alleva E, Lipp H-P, Fasolo A, Nadel L, Ricceri L, eds), pp 37–62. The Netherlands: Kluwer Academic Publishers.

Dunbar RIM (1998) The social brain hypothesis. Evol Anthropol 6:178–190.

Dusser de Barenne JG, Garol HW, McCulloch WS (1941) The 'motor' cortex of the chimpanzee. J Neurophysiol 4:287–304.

Evarts EV (1966) Pyramidal tract activity associated with a conditioned hand movement in the monkey. J Neurophysiol 29:1011–1027.

Evarts EV (1968) Relation of pyramidal tract activity to force exerted during voluntary movement. J Neurophysiol 31:14–27.

Fernández-Carriba S, Loeches A, Morcillo A, Hopkins WD (2002) Asymmetry in facial expression of emotions by chimpanzees. Neuropsychologia 40:1523–1533.

Fleagle JG (1999) Primate Adaptation and Evolution. San Diego: Academic Press.

Foerster O (1936) Motorische Felder und Bahnen. In: Handbuch der Neurologie (Bumke O, Foerster O, eds), pp 1–448. Berlin: Springer.

Fogassi L, Gallese V, Gentilucci M, Luppino G, Matelli M, Rizzolatti G (1994) The fronto-parietal cortex of the prosimian Galago: Patterns of cytochrome oxidase activity and motor maps. Behav Brain Res 60:91–113.

Fox PT, Huang A, Parsons LM, Xiong JH, Zamarippa F, Rainey L, Lancaster JL (2001) Location-probability profiles for the mouth region of human primary motor-sensory cortex: Model and validation. Neuroimage 13:196–209.

Fridlund AJ (1994) Human Facial Expression: An Evolutionary View. San Diego, CA: Academic Press.

Gentilucci M, Fogassi L, Luppino G, Matelli M, Camarda R, Rizzolatti G (1988) Functional organization of inferior area 6 in the macaque monkey. I. Somatotopy and the control of proximal movements. Exp Brain Res 71:475–490.

Geyer S, Schleicher A, Zilles K (1999) Areas 3a, 3b, and 1 of human primary somatosensory cortex. 1. Microstructural organization and interindividual variability. Neuroimage 10:63–83.

Geyer S, Matelli M, Luppino G, Zilles K (2000) Functional neuroanatomy of the primate isocortical motor system. Anat Embryol (Berl) 202:443–474.

Geyer S, Matelli M, Luppino G, Schleicher A, Jansen Y, Palomero-Gallagher N, Zilles K (1998) Receptor autoradiographic mapping of the mesial motor and premotor cortex of the macaque monkey. J Comp Neurol 397:231–250.

Geyer S, Ledberg A, Schleicher A, Kinomura S, Schormann T, Bürgel U, Klingberg T, Larsson J, Zilles K, Roland PE (1996) Two different areas within the primary motor cortex of man. Nature 382:805–807.

Glezer II (1958) Area relationships in the precentral region in a comparative-anatomical series of primates. Arkh Anat 2:26.

Godschalk M, Lemon RN, Kuypers HG, Ronday HK (1984) Cortical afferents and efferents of monkey postarcuate area: an anatomical and electrophysiological study. Exp Brain Res 56: 410–424.

Godschalk M, Mitz AR, van Duin B, van der Burg H (1995) Somatotopy of monkey premotor cortex examined with microstimulation. Neurosci Res 23:269–279.

Goodall J (1986) The Chimpanzees of Gombe. Cambridge, MA: Harvard University Press.

Green HD, Walker AE (1938) The effects of ablation of the cortical motor face area in monkeys. J Neurophysiol 1:262–280.

Groswasser Z, Korn C, Groswasser-Reider I, Solzi P (1988) Mutism associated with buccofacial apraxia and bihemispheric lesions. Brain Lang 34:157–168.

Grünbaum ASF, Sherrington CS (1903–04) Observations on the physiology of the cerebral cortex of the anthropoid apes. Proc Royal Soc 72:152–155.

Harcourt AH (1988) Alliances in contests and social intelligence. In: Machiavellian Intelligence (Byrne RW, Whiten A, eds), pp 132–152. Oxford, UK: Clarendon Press.

Harvey PH, Martin RD, Clutton-Brock TH (1987) Life histories in comparative perspective. In: Primate Societies (Smuts BB, Cheney DL, Seyfarth RM, Wrangham RW, Strusaker TT, eds), pp 181–196. Chicago, IL: University of Chicago Press.

Hast MH, Milojevic B (1966) The response of the vocal folds to electrical stimulation of the inferior frontal cortex of the squirrel monkey. Acta Otolaryngol 61:196–204.

Hast MH, Fischer JM, Wetzel AB, Thompson VE (1974) Cortical motor representation of the laryngeal muscles in *Macaca mulatta*. Brain Res 73:229–240.

Haug R (1987) Brain sizes, surfaces, and neuronal sizes of the cortex cerebri: A stereological investigation of man and his variability and a comparison with some mammals (primates, whales, marsupials, insectivores, and one elephant). Am J Anat 180:126–142.

Hauser MD (1993) Right hemisphere dominance for the production of facial expression in monkeys. Science 261:475–477.

Hauser MD (1996) The Evolution of Communication. Cambridge, MA: MIT Press.

Hauser MD, Akre K (2001) Asymmetries in the timing of facial and vocal expressions by rhesus monkeys: Implications for hemispheric specialization. Anim Behav 61:391–400.

Hines M (1940) Movements elicited from the precentral gyrus of adult chimpanzees by stimulation with sine wave currents. J Neurophysiol 3: 442–466.

Hof PR, Morrison JH (1995) Neurofilament protein defines regional patterns of cortical organization in the macaque monkey visual system: A quantitative immunohistochemical analysis. J Comp Neurol 352:161–186.

Hof PR, Nimchinsky EA (1992) Regional distribution of neurofilament and calcium-binding proteins in the cingulate cortex of the macaque monkey. Cereb Cortex 2:456–467.

Hof PR, Ungerleider LG, Webster MJ, Gattass R, Adams MM, Sailstad CA, Morrison JH (1996) Neurofilament protein is differentially distributed in subpopulations of corticocortical projection neurons in the macaque monkey visual pathways. J Comp Neurol 376:112–127.

Hoffman DS, Luschei ES (1980) Responses of monkey precentral cortical cells during a controlled jaw bite task. J Neurophysiol 44:333–348.

Holloway RL (1968) The evolution of the primate brain: Some aspects of quantitative relations. Brain Res 7:121–172.

Holloway RL (1973) The role of human social behavior in the evolution of the brain. New York: American Museum of Natural History.

Huang CS, Hiraba H, Sessle BJ (1989) Input-output relationships of the primary face motor cortex in the monkey *(Macaca fascicularis)*. J Neurophysiol 61:350–362.

Huang CS, Sirisko MA, Hiraba H, Murray GM, Sessle BJ (1988) Organization of the primate face motor cortex as revealed by intracortical microstimulation and electrophysiological identification of afferent inputs and corticobulbar projections. J Neurophysiol 59:796–818.

Humphrey DR (1986) Representation of movements and muscles within the primate precentral motor cortex: Historical and current perspectives. Fed Proc 45:2687–2699.

Humphrey N (1976) The social function of intellect. In: Growing Points in Ethology (Bateson PG, Hinde RA, eds), pp 303–317. Cambridge, UK: Cambridge University Press.

Ichikawa M, Arissian K, Asanuma H (1985) Distribution of corticocortical and thalamocortical synapses on identified motor cortical neurons in the cat: Golgi, electron microscopic and degeneration study. Brain Res 345:87–101.

Ilinsky IA, Jouandet ML, Goldman-Rakic PS (1985) Organization of the nigrothalamocortical system in the rhesus monkey. J Comp Neurol 236:315–330.

Iwase M, Ouchi Y, Okada H, Yokoyama C, Nobezawa S, Yoshikawa E, Tsukada H, Takeda M, Yamashita K, Yamaguti K, Kuratsune H, Shimizu A, Watanabe Y (2002) Neural substrates of human facial expression of pleasant emotion induced by comic films: A PET study. Neuroimage 17:758–768.

Jones EG (1985) The Thalamus. New York: Plenum Press.

Jones EG, Powell TP (1968) The ipsilateral cortical connexions of the somatic sensory areas in the cat. Brain Res 9:71–94.

Jones EG, Wise SP (1977) Size, laminar and columnar distribution of efferent cells in the sensory-motor cortex of monkeys. J Comp Neurol 175: 391–438.

Jones EG, Coulter JD, Hendry SH (1978) Intracortical connectivity of architectonic fields in the somatic sensory, motor and parietal cortex of monkeys. J Comp Neurol 181:291–347.

Jürgens U, Kirzinger A, von Cramon D (1982) The effects of deep-reaching lesions in the cortical face area on phonation: A combined case report and experimental monkey study. Cortex 18:125–140.

Kaneko T, Caria MA, Asanuma H (1994) Information processing within the motor cortex. II. Intracortical connections between neurons receiving somatosensory input and motor output neurons of the cortex. J Comp Neurol 345: 161–171.

Kimura D (1993) Neuromotor Mechanisms in Human Communication. New York: Oxford University Press.

Kirzinger A, Jürgens U (1982) Cortical lesion effects and vocalization in the squirrel monkey. Brain Res 233:299–315.

Kleim JA, Lussnig E, Schwarz ER, Comery TA, Greenough WT (1996) Synaptogenesis and Fos expression in the motor cortex of the adult rat after motor skill learning. J Neurosci 16:4529–4535.

Kretschmann HJ, Tafesse U, Herrmann A (1982) Different volume changes of cerebral cortex and white matter during histological preparation. Microsc Acta 86:13–24.

Künzle H (1978) An autoradiographic analysis of the efferent connections from premotor and adjacent prefrontal regions (areas 6 and 9) in *Macaca fascicularis*. Brain Behav Evol 15:185–234.

Kwan HC, MacKay WA, Murphy JT, Wong YC (1978) Spatial organization of precentral cortex in awake primates. II. Motor outputs. J Neurophysiol 41:1120–1131.

Lassek AM (1940) The human pyramidal tract. II. A numerical investigation of the Betz cells of the motor area. Arch Neurol Psychiat 44:718–724.

Lassek AM (1941) The pyramidal tract of the monkey. Betz cells and pyramidal tract enumeration. J Comp Neurol 74:193–202.

Lauer EW (1952) Ipsilateral facial representation in motor cortex of macaque. J Neurophysiol 15:1–4.

Leichnetz GR (1986) Afferent and efferent connections of the dorsolateral precentral gyrus (area 4, hand/arm region) in the macaque monkey, with comparisons to area 8. J Comp Neurol 254:460–492.

Leinonen L, Nyman G (1979) Functional properties of cells in the anterolateral part of area 7 associative face area of awake monkeys. Exp Brain Res 34:321–333.

Leyton ASF, Sherrington CS (1917) Observations on the excitable cortex of the chimpanzee, orang-utan, and gorilla. Q J Exp Psychol 11: 137–222.

Lin LD, Murray GM, Sessle BJ (1994a) Functional properties of single neurons in the primate face primary somatosensory cortex. I. Relations with trained orofacial motor behaviors. J Neurophysiol 71:2377–2390.

Lin LD, Murray GM, Sessle BJ (1994b) Functional properties of single neurons in the primate face primary somatosensory cortex. II. Relations with different directions of trained tongue protrusion. J Neurophysiol 71:2391–2400.

Lotze M, Erb M, Flor H, Huelsmann E, Godde B, Grodd W (2000) fMRI evaluation of somatotopic representation in human primary motor cortex. Neuroimage 11:473–481.

Luppino G, Matelli M, Camarda R, Rizzolatti G (1993) Corticocortical connections of area F3 (SMA-proper) and area F6 (pre-SMA) in the macaque monkey. J Comp Neurol 338:114–140.

Mao C-C, Coull BM, Gopler LAC, Rau MT (1989) Anterior operculum syndrome. Neurology 39: 1169–1172.

Maple TL (1980) Orang-utan Behavior. New York: Van Nostrand, Reinhold Company.

Mariani C, Spinnler H, Sterzi R, Vallar G (1980) Bilateral perisylvian softening: Bilateral anterior opercular syndrome (Foix-Chavany-Marie syndrome). J Neurol 223:269–284.

Martin RD (1981) Relative brain size and basal metabolic rate in terrestrial vertebrates. Nature 293:57–60.

Matelli M, Luppino G, Rizzolatti G (1985) Patterns of cytochrome oxidase activity in the frontal agranular cortex of the macaque monkey. Behav Brain Res 18:125–136.

Matelli M, Luppino G, Rizzolatti G (1991) Architecture of superior and mesial area 6 and the adjacent cingulate cortex in the macaque monkey. J Comp Neurol 311:445–462.

Matelli M, Luppino G, Fogassi L, Rizzolatti G (1989) Thalamic input to inferior area 6 and area 4 in the macaque monkey. J Comp Neurol 280:468–488.

Mauss T (1908) Die faserarchitektonische Gliederung der Grosshirnrinde bei den niederen Affen. J Psychol Neurol 13:263–325.

Mauss T (1911) Die faserarchitektonische Gliederung des Cortex cerebri von den anthropomorphen Affen. J Psychol Neurol 18:410–467.

Mayer O (1912) Mikrometrische Untersuchungen über Zelldichtigkeit der Grosshirnrinde bei den Affen. Jahrb Psychol u Neurol 17:233–251.

McGuinness E, Sivertsen D, Allman JM (1980) Organization of the face representation in macaque motor cortex. J Comp Neurol 193:591–608.

Meyer G (1987) Forms and spatial arrangement of neurons in the primary motor cortex of man. J Comp Neurol 262:402–428.

Morecraft RJ, Van Hoesen GW (1992) Cingulate input to the primary and supplementary motor cortices in the rhesus monkey: Evidence for somatotopy in areas 24c and 23c. J Comp Neurol 322:471–489.

Morecraft RJ, Van Hoesen GW (1993) Frontal granular cortex input to the cingulate (M3), supplementary (M2) and primary (M1) motor cortices in the rhesus monkey. J Comp Neurol 337:669–689.

Morecraft RJ, Schroeder CM, Keifer J (1996) Organization of face representation in the cingulate cortex of the rhesus monkey. NeuroReport 7: 1343–1348.

Muakkassa KF, Strick PL (1979) Frontal lobe inputs to primate motor cortex: Evidence for four somatotopically organized ‘premotor’ areas. Brain Res 177:176–182.

Murray GM, Sessle BJ (1992a) Functional properties of single neurons in the face primary motor cortex of the primate. I. Input and output features of tongue motor cortex. J Neurophysiol 67:747–758.

Murray GM, Sessle BJ (1992b) Functional properties of single neurons in the face primary motor cortex of the primate. III. Relations with different directions of trained tongue protrusion. J Neurophysiol 67:775–785.

Murray GM, Sessle BJ (1992c) Functional properties of single neurons in the face primary motor cortex of the primate. II. Relations with trained orofacial motor behavior. J Neurophysiol 67: 759–774.

Murray GM, Lin LD, Moustafa EM, Sessle BJ (1991) Effects of reversible inactivation by cooling of the primate face motor cortex on the performance of a trained tongue-protrusion task and a trained biting task. J Neurophysiol 65:511–530.

Myers RE (1969) Neurology of social communication in primates. In: Neurology, Physiology, and Infectious Diseases (Hofer HO, ed), pp 1–9. Basel: Karger.

Myers RE (1976) Comparative neurology of vocalization and speech: Proof of a dichotomy. Ann N Y Acad Sci 280:745–760.

Narita N, Yamamura K, Yao D, Martin RE, Sessle BJ (1999) Effects of functional disruption of lateral pericentral cerebral cortex on primate swallowing. Brain Res 824:140–145.

Narita N, Yamamura K, Yao D, Martin RE, Masuda Y, Sessle BJ (2002) Effects on mastication of reversible bilateral inactivation of the lateral pericentral cortex in the monkey *(Macaca fascicularis)*. Arch Oral Biol 47:673–688.

O'Malley RC, McGrew WC (2000) Oral tool use by captive orangutans *(Pongo pygmaeus)*. Folia Primatol 71:334–341.

Pandya DN, Vignolo LA (1971) Intra- and interhemispheric projections of the precentral, premotor and arcuate areas in the rhesus monkey. Brain Res 26:217–233.

Pannese E (1994) Neurocytology: Fine Structure of Neurons, Nerve Processes, and Neuroglial Cells. New York: Thieme Medical Publisher, Inc.

Pavlides C, Miyashita E, Asanuma H (1993) Projection from the sensory to the motor cortex is important in learning motor skills in the monkey. J Neurophysiol 70:733–741.

Penfield W, Boldrey E (1937) Somatic motor and sensory representation in the cerebral cortex of man studied by electrical stimulation. Brain 60:389–443.

Penfield W, Rasmussen T (1950) The Cerebral Cortex of Man: A Clinical Study of Localization of Function. New York: Macmillan.

Peters M, Ploog D (1973) Communication among primates. Ann Rev Physiol 35:221–242.

Porter LL, Sakamoto K (1988) Organization and synaptic relationships of the projection from the primary sensory to the primary motor cortex in the cat. J Comp Neurol 271:387–396.

Preuschoft S, van Hooff JA (1995) Homologizing primate facial displays: A critical review of methods. Folia Primatol (Basel) 65:121–137.

Preuss TM, Stepniewska I, Kaas JH (1996) Movement representation in the dorsal and ventral premotor areas of owl monkeys: A microstimulation study. J Comp Neurol 371:649–676.

Preuss TM, Stepniewska I, Jain N, Kaas JH (1997) Multiple divisions of macaque precentral motor cortex identified with neurofilament antibody SMI-32. Brain Res 767:148–153.

Prothero J (1997) Scaling of cortical neuron density and white matter volume in mammals. J Hirnforsch 38:513–524.

Rademacher J, Caviness VS, Steinmetz H, Galaburda A (1993) Topographical variations of the human primary cortices: Implications for neuroimaging, brain mapping and neurobiology. Cereb Cortex 3:313–329.

Rademacher J, Bürgel U, Geyer S, Schormann T, Schleicher A, Freund HJ, Zilles K (2001) Variability and asymmetry in the human precentral motor system. A cytoarchitectonic and myeloarchitectonic brain mapping study. Brain 124: 2232–2258.

Ringo JL (1991) Neuronal interconnection as a function of brain size. Brain Behav Evol 38:1–6.

Rinn WE (1984) The neuropsychology of facial expression: A review of the neurological and psychological mechanisms for producing facial expressions. Psychol Bull 95:52–77.

Rioult-Pedotti MS, Friedman DP, Hess G, Donoghue JP (1998) Strengthening of horizontal cortical connections following skill learning. Nature Neurosci 1:230–234.

Rivara C-B, Sherwood CC, Bouras C, Hof PR (2003) Stereologic characterization and spatial distribution patterns of Betz cells in human primary motor cortex. Anat Rec 270A:137–151.

Salmelin R, Sams M (2002) Motor cortex involvement during verbal versus non-verbal lip and tongue movements. Hum Brain Mapp 16:81–91.

Samulack DD, Waters RS, Dykes RW, McKinley PA (1990) Absence of responses to microstimulation at the hand-face border in baboon primary motor cortex. Can J Neurol Sci 17:24–29.

Sanides F (1970) Functional architecture of motor and sensory cortices in primates in the light of a new concept of neocortex evolution. In: The Primate Brain: Advances in Primatology, Volume 1 (Noback CR, Montagna W, eds), pp 209–224. New York, NY: Appleton-Century-Crofts.

Schlaug G, Armstrong E, Schleicher A, Zilles K (1993) Layer V pyramidal cells ion the adult human cingulate cortex: A quantitative Golgi study. Anat Embryol 187:515–522.

Schleicher A, Zilles K (1990) A quantitative approach to cytoarchitectonics: Analysis of structural inhomogeneities in nervous tissue using an image analyzer. J Microsc 157:367–381.
Schleicher A, Amunts K, Geyer S, Kowalski T, Schormann T, Palomero-Gallagher N, Zilles K (2000) A stereological approach to human cortical architecture: Identification and delineation of cortical areas. J Chem Neuroanat 20: 31–47.
Schultz A (1969) The Life of Primates. London: Weidenfeld and Nicholson.
Semendeferi K, Armstrong E, Schleicher A, Zilles K, Van Hoesen GW (1998) Limbic frontal cortex in hominoids: A comparative study of area 13. Am J Phys Anthropol 106:129–155.
Semendeferi K, Armstrong E, Schleicher A, Zilles K, Van Hoesen GW (2001) Prefrontal cortex in humans and apes: a comparative study of area 10. Am J Phys Anthropol 114:224–241.
Shariff GA (1953) Cell counts in the primate cerebral cortex. J Comp Neurol 98:381–400.
Sherwood CC, Broadfield DC, Holloway RL, Gannon PJ, Hof PR (2003a) Variability of Broca's area homologue in African great apes: Implications for language evolution. Anat Rec 271A:276–285.
Sherwood CC, Lee PH, Rivara C-B, Holloway RL, Gilissen EPE, Simmons RMT, Hakeem A, Allman JM, Erwin JM, Hof PR (2003b) Evolution of specialized pyramidal neurons in primate visual and motor cortex. Brain Behav Evol 61: 28–44.
Sherwood CC, Holloway RL, Erwin JM, Hof PR (2004) Cortical orofacial motor representation in old world monkeys, great apes, and humans. II. Stereologic analysis of chemoarchitecture. Brain Behav Evol 63:82–106.
Sloper JJ (1973) An electron microscope study of the termination of afferent connections to the primate motor cortex. J Neurocytol 2:361–368.
Sloper JJ, Powell TP (1979) An experimental electron microscopic study of afferent connections to the primate motor and somatic sensory cortices. Phil Trans R Soc Lond B Biol Sci 285: 199–226.
Stepanyants A, Hof PR, Chklovskii DB (2002) Geometry and structural plasticity of synaptic connections. Neuron 34:275–288.
Stephan H, Frahm HD, Baron G (1981) New and revised data on volumes of brain structures in insectivores and primates. Folia Primatol 35: 1–29.
Stepniewska I, Preuss TM, Kaas JH (1993) Architectonics, somatotopic organization, and ipsilateral cortical connections of the primary motor area (M1) of owl monkeys. J Comp Neurol 330:238–271.
Strick PL (1973) Light microscopic analysis of the cortical projection of the thalamic ventrolateral nucleus in the cat. Brain Res 55:1–24.
Strick PL, Sterling P (1974) Synaptic termination of afferents from the ventrolateral nucleus of the thalamus in the cat motor cortex. A light and electron microscopy study. J Comp Neurol 153:77–106.
Sutton D, Larson C, Lindeman RC (1974) Neocortical and limbic lesion effects on primate phonation. Brain Res 71:61–75.
Svetukhina VM (1956) Cytoachitectonics of the precoronal and frontal cortex in the Carnivora. In. Moscow, Dissertation.
Szentágothai J (1983) The modular architectonic principle of neuronal centers. Rev Physiol Biochem Pharmacol 98:11–61.
Tigges J, Herndon JG, Peters A (1990) Neuronal population of area 4 during the life span of the rhesus monkey. Neurobiol Aging 11:201–208.
Tokuno H, Takada M, Nambu A, Inase M (1997) Reevaluation of ipsilateral corticocortical inputs to the orofacial region of the primary motor cortex in the macaque monkey. J Comp Neurol 389:34–48.
Tower DB (1954) Structural and functional organization of mammalian cerebral cortex: The correlation of neurone density with brain size. J Comp Neurol 101:19–52.
Tower DB, Young OM (1973a) Interspecies correlations of cerebral cortical oxygen consumption, acetylcholinesterase activity and chloride content: Studies on the brains of the fin whale *(Balaenoptera physalus)* and the sperm whale *(Physeter catadon)*. J Neurochem 20:253–267.
Tower DB, Young OM (1973b) The activities of butyrylcholinesterase and carbonic anhydrase, the rate of anaerobic glycolysis and the question of constant density of glial cells in cerebral cortices of mammalian species from mouse to whale. J Neurochem 20:269–278.
van Hooff JARAM (1962) Facial expressions in higher primates. Symp Zool Soc Lond 8:97–125.
van Hooff JARAM (1967) The facial displays of catarrhine monkeys and apes. In: Primate Ethology (Morris D, ed), pp 7–68. Chicago: Aldine.
van Lawick-Goodall J (1968) A preliminary report on expressive movements and communication in the Gombe Stream Chimpanzees. In: Primates (Jay PC, ed), pp 313–374. New York: Holt, Reinhart and Winston.
van Schaik CP, Fox EA, Sitompul AF (1996) Manufacture and use of tools in wild Sumatran orangutans. Implications for human evolution. Naturwissenschaften 83:186–188.
Vogt BA, Pandya DN (1978) Cortico-cortical connections of somatic sensory cortex (areas 3, 1 and 2) in the rhesus monkey. J Comp Neurol 177:179–191.
Vogt C, Vogt O (1907) Zur Kenntnis der elektrisch erregbaren Hirnrindengebiete bei den Säugetieren. J Psychol Neurol 8:277.
Vogt C, Vogt O (1919) Allgemeinere Ergebnisse unserer Hirnforschung. J Psychol Neurol 25: 279–461.
von Bonin G (1949) Architecture of the precentral motor cortex and some adjacent areas. In: The Precentral Motor Cortex (Bucy PC, ed), pp 7–82. Urbana, IL: University of Illinois Press.
von Bonin G, Bailey P (1947) The Neocortex of *Macaca mulatta*. Urbana, IL: University of Illinois Press.
von Economo C, Koskinas GN (1925) Die Cytoarchitektonik der Hirnrinde des erwachsenen Menschen. Wien und Berlin: J. Springer.
Walker AE (1938) Thalamus of the chimpanzee. IV. Thalamic projections to the cerebral cortex. J Anat 73:37–93.
Walker AE, Green HD (1938) Electrical excitability of the motor face area: A comparative study in primates. J Neurophysiol 1:152–165.
Waters RS, Samulack DD, Dykes RW, McKinley PA (1990) Topographic organization of baboon primary motor cortex: Face, hand, forelimb, and shoulder representation. Somatosens Mot Res 7:485–514.
White LE, Andrews TJ, Hulette C, Richards A, Groelle M, Paydarfar J, Purves D (1997) Structure of the human sensorimotor system. I: Morphology and cytoarchitecture of the central sulcus. Cereb Cortex 7:18–30.
Whiten A, Byrne RW (1988) Tactical deception in primates. Behav Brain Sci 11:233–244.
Wise SP (1985) The primate premotor cortex: Past, present and preparatory. Ann Rev Neurosci 8: 1–19.
Withers GS, Greenough WT (1989) Reach training selectively alters dendritic branching in subpopulations of layer II–III pyramids in rat motor-somatosensory forelimb cortex. Neuropsychologia 27:61–69.
Woolsey CN (1958) Organization of somatic sensory and motor areas of the cerebral cortex. In: Biological and Biochemical Bases of Behavior (Harlow HF, Woolsey CN, eds), pp 63–81. Madison, WI: University of Wisconsin Press.
Wree A, Schleicher A, Zilles K (1982) Estimation of volume fractions in nervous tissue with an image analyzer. J Neurosci Methods 6:29–43.
Yamamura K, Narita N, Yao D, Martin RE, Masuda Y, Sessle BJ (2002) Effects of reversible bilateral inactivation of face primary motor cortex on mastication and swallowing. Brain Res 944:40–55.
Yao D, Yamamura K, Narita N, Murray GM, Sessle BJ (2002a) Effects of reversible cold block of face primary somatosensory cortex on orofacial movements and related face primary motor cortex neuronal activity. Somatosens Mot Res 19:261–271.
Yao D, Yamamura K, Narita N, Martin RE, Murray GM, Sessle BJ (2002b) Neuronal activity patterns in primate primary motor cortex related to trained or semiautomatic jaw and tongue movements. J Neurophysiol 87:2531–2541.
Yerkes RM, Yerkes AW (1929) The Great Apes. New Haven, CT: Yale University Press.
Yousry TA, Schmid UD, Alkadhi H, Schmidt D, Peraud A, Buettner A, Winkler P (1997) Localization of the motor hand area to a knob on the precentral gyrus. A new landmark. Brain 120: 141–157.
Yumiya H, Ghez C (1984) Specialized subregions in the cat motor cortex: Anatomical demonstration of differential projections to rostral and caudal sectors. Exp Brain Res 53:259–276.
Zilles K (1990) Cortex. In: The Human Nervous System (Paxinos G, ed), pp 757–852. San Diego, CA: Academic Press.
Zilles K, Rehkämper G (1988) The brain, with special reference to the telencephalon. In: Orangutan Biology (Schwartz JH, ed), pp 157–176. New York: Oxford University Press.
Zilles K, Stephan H, Schleicher A (1982) Quantitative cytoarchitectonics of the cerebral cortices of several prosimian species. In: Primate Brain Evolution: Methods and Concepts (Armstrong E, Falk D, eds), pp 177–202. New York: Plenum Press.